

CHINESE INDUSTRIAL POLICY AND INTERNATIONAL SPILLOVERS: EVIDENCE FROM 70,000 POLICY DOCUMENTS*

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May 18, 2026

Abstract

We study the international spillovers of Chinese industrial policy. We measure policy support at the sector-province-year level from official government documents, using supervised classification of policy sentences across industries and over time. The measure captures local implementation of national priorities and predicts realized subsidy receipts in firm accounts. We then combine the policy index with Chinese firm data, customs data, and French administrative data. A variable-markup trade model guides the empirical design: policy support can lower Chinese firms' marginal costs or improve market access, while foreign competitors are displaced when Chinese policy tightens competition in markets where China is already present. The evidence supports this mechanism. Targeted Chinese firms receive more subsidies, expand in export markets, and lower quality-adjusted export prices, without a detectable increase in firm-level markups. Policy support is therefore passed through to foreign markets rather than retained as higher price-cost margins. These domestic responses generate spillovers abroad. French firms lose export revenue in products more exposed to Chinese policy, and firm-level production falls among exporters whose pre-period portfolios were concentrated in markets where China was an important competitor. The results show that industrial policy propagates internationally through product-market competition.

JEL codes: F12, F13, L52, O25, C81.

Keywords: Industrial Policy, China, Variable Markups, International Spillovers, Text-as-Data.

*The authors are grateful to Lily Tong for her invaluable assistance in developing the methods, and to the Bright Initiative for providing parts of the technical infrastructure for the Reading China Better project.

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1 Introduction

China’s industrial policy is a domestic intervention with international incidence. It is designed and implemented inside China, but it changes the competitive environment faced by firms abroad. This incidence is difficult to study because the relevant policy variation is hard to observe. Conservative estimates, expressed in U.S. dollars, place Chinese industrial subsidies at \$350 billion in 2019, compared with \$84 billion in the United States (DiPippo et al., 2022). Yet these transfers are only one part of a broader policy system. Chinese industrial policy spans many sectors, combines fiscal, credit, regulatory, procurement, and infrastructure instruments, and is implemented through a decentralized system in which provincial governments translate central directives into local commitments. National plans reveal broad priorities, but they do not measure local implementation. WTO notifications are incomplete, firm-level subsidy data are not public, and existing studies typically focus on one instrument or one sector (Kalouptsi, 2018; Barwick et al., 2024; Bai et al., 2019; Branstetter et al., 2024). This leaves open a central question: how does Chinese industrial policy affect domestic firms and foreign competitors through international product markets?

This paper answers this question by measuring local policy implementation and tracing its effects across firms and countries. We construct sector-province-year measures of Chinese industrial policy from 70,000 official government documents in the Reading China Better database (Brussee, 2025). A BERT-based classifier trained on 5,000 expert-annotated sentences identifies whether a policy sentence provides financial support, promotes a sector, or restricts activity. The resulting index covers 83 industries and all provinces from 2011 to 2024. It measures documented policy intensity, not realized expenditure. This distinction is central. Policy documents do not provide a fiscal ledger, but they reveal which sectors governments choose to support, when they support them, and where this support is implemented. The index predicts realized subsidy receipts in firm accounts and rises around the onset of provincial targeting. It therefore captures policy priorities that are informative

about fiscal effort and about non-fiscal instruments that do not appear in subsidy accounts.

The empirical object is the transmission of these policy priorities through trade. We build on [Melitz and Ottaviano \(2008\)](#), with heterogeneous firms, endogenous markups, and export participation. Industrial policy can lower marginal costs or reduce destination fixed costs for targeted Chinese firms. These shocks raise Chinese export activity and affect prices. With variable markups, the price response depends on both the direct cost reduction and the induced change in market toughness. When policy-induced cost reductions pass through to export prices, foreign competitors face lower residual demand in markets where Chinese firms are active. The model delivers the exposure measure used in the French firm analysis: a firm is more exposed when its pre-period export revenue is concentrated in product-destination markets where China is an important competitor and where Chinese policy support is stronger.

The empirical analysis proceeds in three steps. First, we validate the policy measure. Documented policy intensity predicts realized production subsidies at the sector-province-year level, conditional on province-year fixed effects. In an event-study design based on staggered provincial targeting, firm-level subsidies rise after a province begins targeting a sector, while pre-treatment coefficients are close to zero. These results show that the text-based measure is not only policy rhetoric. It contains information about subsequent policy effort.

Second, we estimate the effects of policy exposure on Chinese firms. Targeted firms receive more subsidies, expand exports, and serve more destinations. They also lower quality-adjusted export prices, especially for final goods. Firm-level markups, estimated following [De Loecker and Warzynski \(2012\)](#) and [Akerberg et al. \(2015\)](#), do not rise in event-study estimates. Taken together, these results point to a price-competition mechanism. Policy support appears to improve Chinese firms' competitiveness and to pass through to foreign markets, rather than being retained mainly as higher price-cost margins.

Third, we estimate the international spillovers of Chinese policy using French customs

and balance-sheet data from 2011 to 2020. We assign Chinese policy exposure to each French firm-product export flow and absorb firm-year shocks. Identification therefore comes from comparing products with different exposure to Chinese policy within the same French firm-year. French firms lose export revenue in products more exposed to Chinese policy. At the firm level, exposure also lowers production, with stronger effects when the Chinese policy measure loads on fiscal-support language. These estimates identify displacement through product markets rather than a broad firm-level shock: the same French firm loses more in products more exposed to Chinese policy.

The paper contributes to three literatures. First, it advances the measurement of industrial policy at scale. [Juhász et al. \(2024\)](#) review this literature; [Evenett and Fritz \(2022\)](#) and [Juhász et al. \(2025\)](#) provide cross-country evidence using the Global Trade Alert; and [Rotunno and Ruta \(2024\)](#) study the trade and input-output implications of trade-distortive interventions. For China, text-based work uses policy documents to study industrial-policy design ([Fang et al., 2025](#)), renewable-energy policy ([Gerarden et al., 2026](#)), credit allocation ([Xu et al., 2024](#)), Five-Year Plans ([Chen et al., 2017](#); [Cen et al., 2022](#)), and WTO accession ([Tan, 2021](#)). We construct a validated subnational-sector measure designed for firm-level and trade-spillover analysis, building on text-as-data methods in economics and political science ([Gentzkow et al., 2019](#); [Grimmer et al., 2022](#)). The key object is not national policy intent, but local implementation by sector, province, and year.

Second, the paper contributes to work on the effects of Chinese industrial policy ([Aghion et al., 2015](#); [Chen and Steinwender, 2023](#); [Barwick et al., 2024](#); [Bernard, 2025](#)), including studies of VAT rebates ([Bai et al., 2019](#); [Liu et al., 2019](#)), R&D tax credits ([Chen et al., 2021](#)), export restrictions ([Gourdon et al., 2016](#); [Garred, 2018](#); [Alfaro and Chor, 2025](#)), targeted sectors ([Kalouptsi, 2018](#)), and the historical role of subsidies in industrial development ([Juhász and Steinwender, 2024](#)). It also relates to broader work on trade policy, place-based support, and welfare criteria for targeted interventions ([Criscuolo et al., 2019](#); [Harrison and Rodriguez-Clare, 2010](#); [Goldberg and Pavcnik, 2016](#); [Bartelme and Costinot, 2025](#)). Relative

to studies of a single instrument or sector, we recover firm responses to the joint policy mix and use markups and quality-adjusted prices to distinguish rent capture from pass-through.

Third, the paper contributes to the literature on international spillovers from industrial policy (Rotunno and Ruta, 2024; Lane, 2024; Wu, 2025). Industrial policy can affect foreign firms even when it is implemented entirely at home. The mechanism in this paper is direct product-market competition. Chinese policy support lowers the effective cost of competing in foreign markets; foreign firms lose sales where their product portfolios overlap with China. By combining Chinese policy documents, Chinese firm outcomes, export prices, and French administrative data, we trace this chain from local policy implementation to foreign-firm displacement.

The remainder of the paper is organized as follows. Section 2 describes the institutional background. Section 3 presents the data and measurement strategy. Section 4 documents key empirical facts. Section 5 develops the theoretical framework. Section 6 estimates the effects on Chinese firms. Section 7 traces the international spillovers to French firms. Section 8 concludes.

2 Institutional background

The institutional setting determines both the object we measure and the source of identifying variation. Chinese industrial policy is centrally guided but locally implemented. Central authorities set broad priorities, while provincial and municipal governments choose instruments, allocate fiscal resources, and adapt directives to local economic objectives. This structure implies that national documents are informative about strategic direction, but that implementation varies across provinces, sectors, and years.

Decentralization and local incentives. China operates under “regionally decentralized authoritarianism” (Xu, 2011). Subnational governments implement central priorities and control most public spending: local governments account for 86% of public expenditure in 2023 (National Bureau of Statistics of China, 2023). Provincial governments also have

substantial authority over industrial development, infrastructure, investment promotion, and fiscal allocation. Central and provincial policies therefore interact in a system that is neither purely top-down nor fully decentralized. National plans define priority sectors. Provinces decide how strongly, and when, to support those sectors.

This allocation of authority creates competition across provinces. Provincial leaders face promotion incentives tied to local economic performance, and developing flagship industries raises both fiscal revenue and political visibility (Xu, 2011). Provinces therefore compete to attract firms, investment, and sectoral rents through subsidies, tax preferences, credit support, procurement, infrastructure, and other forms of coordinated assistance. The logic is close to the “race to the top” emphasized by Aghion et al. (2015). For our purposes, the implication is empirical: local incentives generate variation in the timing and intensity of sectoral support across provinces.

Policy instruments and transmission. Chinese industrial policy uses a broad mix of instruments. Official documents announce direct subsidies, tax expenditures, special funds, credit support, land and infrastructure provision, procurement preferences, and sector-specific regulatory support. These instruments are commonly issued through opinions and planning documents by provincial or municipal People’s Governments. They are the documents from which we construct the policy-intensity measures in Section 3.

The documents also contain concrete sectoral commitments. For example, the 2020 Ministry of Finance notice on new-energy vehicles revised financial subsidies for NEV promotion and application, while data-centre policy appears in national planning documents such as the 2021 Fourteenth Five-Year Plan and subsequent MIIT implementation notices. These examples illustrate the object measured below: documented support for particular sectors, issued through formal policy channels.

The documentary channel matters because central directives do not map mechanically into local policy. Local governments reformulate higher-level priorities according to local mandates and resources (Ma, 2022; Birney, 2014). They also continue to experiment under

the more centralized regime of Xi Jinping (Heilmann, 2018; Tong, 2024). Central directives often use deliberate ambiguity to preserve political control while allowing adaptation, a pattern Ang (2024) calls “adaptive policy communication.” As a result, the same sector can be targeted in different provinces at different times and with different intensity. This staggered local implementation is the within-country variation exploited below.

The timing of Chinese industrial policy also matters for the sample period. Before 2005, policy was largely horizontal and focused on market reforms rather than sectoral targeting. The Medium- and Long-Term Program of Science and Technology in 2006 and the Strategic Emerging Industries initiative in 2010 marked a shift toward explicit sectoral support. After 2015, Made in China 2025 coordinated support across subsidies, tax benefits, credit allocation, and procurement preferences and directed it toward semiconductors, electric vehicles, biomedicine, and advanced manufacturing (Naughton, 2021). This evolution helps explain why policy intensity rises over time in our data and why the post-2015 period is especially informative.

Scope of the document-based measure. Policy documents measure stated priorities and announced instruments. They do not provide a continuous ledger of realized fiscal transfers. A province may target a sector in 2012 and continue supporting it in later years without issuing new documents. Our measure therefore captures policy *onset* and *intensification*, rather than the stock of ongoing support. The event-study design in Section 4 follows this interpretation by measuring outcomes relative to the first year in which a province targets a sector and allowing effects to persist thereafter.

The measure also does not exhaust all channels of state support. State-owned enterprises (SOEs) receive resources through state budgets, state-owned banks, directed investment, and ownership-based coordination. These channels are only partly reflected in published policy texts. SOEs also play a steering role in capital-intensive sectors, including shipbuilding, energy, and logistics, where state ownership can substitute for explicit subsidy. Our estimates should therefore be interpreted as the effects of documented policy intensity, which may also

proxy for unobserved credit and ownership channels. If undocumented support is positively correlated with documented priorities, the index understates total state support.

Subsidy opacity reinforces the value of a text-based measure. Chinese subsidies are not fully observed in public fiscal data, and firm-level subsidy information is not systematically disclosed. Conservative estimates, expressed in U.S. dollars, place Chinese industrial subsidies at \$350 billion in 2019, compared with \$84 billion for the United States (DiPippo et al., 2022); the lack of firm-level transparency has been emphasized by García-Herrero and Schindowski (2025). This opacity complicates direct measurement and WTO notification. Our approach does not recover realized fiscal transfers. It measures the intensity of subsidizing intent expressed in official documents. The validation exercise in Section 4 shows that this intent predicts realized firm-level subsidy receipts.

Implications for empirical design. The institutional structure has two empirical implications. First, central-government documents alone are insufficient. They identify national priorities but miss the local implementation margin that affects firms. We therefore use both national and provincial documents, and we rely primarily on provincial variation for identification. Second, national policy targets nearly all sectors at least once and provides little credible control variation. Provincial targeting is staggered across sectors and regions. This staggered implementation supplies the variation used to estimate the effects of Chinese industrial policy on Chinese firms and, through product-destination exposure, on foreign competitors.

3 Data and measurement

This section describes how we construct the industrial-policy index and the firm-level datasets used in the empirical analysis. The measurement goal is to recover variation in documented policy support at the sector-province-year level. The institutional background in Section 2 implies that this is the relevant level for local implementation. The empirical sections then link this measure to Chinese firm outcomes, Chinese export behavior, and exposure of French

firms to Chinese competition.

3.1 From documents to measures

The construction addresses three measurement problems. First, government transparency varies over time, across document types, and across regions. We therefore restrict the sample to years and document categories with relatively stable disclosure. Second, policy documents combine concrete fiscal commitments with broader promotional language. We classify sentence-level policy attitudes using supervised machine learning trained on expert-coded examples. Third, text-based measures may depend on classification choices. We therefore validate the classifier directly and compare the resulting index with a dictionary-based alternative. Appendix A provides full documentation.

Data source. We use the *Reading China Better* (RCB) database, which collects official and quasi-official documents from 139 government websites (Brussee, 2025). RCB covers national, ministerial, provincial, and selected prefectural authorities and provides the original Chinese text and document metadata. These documents are released under China’s Open Government Information (OGI) initiative, introduced nationwide in 2008. The OGI framework mandates publication of laws, regulations, and policy directives. Its stated objectives are administrative: improving compliance, reducing principal-agent frictions, and signaling adherence to international obligations (Chen et al., 2022; Horsley, 2007).

Relative to alternatives such as the *Global Trade Alert*, RCB provides the full text of original documents in Chinese rather than English-language summaries prepared by editors, and covers multiple levels of government, which is essential in a decentralized policy system.¹

Mitigating missingness. A central challenge is partial disclosure. We take three steps. First, we restrict the sample to 2011–2024, beginning with the 12th Five-Year Plan, when disclosure rates stabilized. Second, we restrict attention to document types with reliable

¹For example, compare the full [Henan provincial directive on artificial intelligence](#) with its [GTA summary](#). GTA provides a useful synopsis but omits details contained in the original. NLP methods applied to summaries may capture editorial phrasing rather than underlying policy content.

disclosure: *opinions* (*yijian*, 意见) and planning documents, *guihua* (规划), *fang'an* (方案), *jihua* (计划), and *gangyao* (纲要). These types are consistently published, and transparency depends more on document type than on topic. The restrictions yield 69,388 documents. Third, we normalize sectoral references by total document length in each region-year, so that variation in disclosure volume does not mechanically generate variation in measured policy intensity. Appendix B documents the underlying transparency patterns.

Classification and aggregation. We construct the policy index in two steps. First, following Cen et al. (2022), we identify references to economic sectors using the China Securities Regulatory Commission industry classification. This procedure yields approximately 700,000 sentences referencing 83 sectors. We match these sectors to the Chinese Industrial Classification at the 2-digit level.

Second, we classify the policy attitude expressed toward each sector using a BERT-based model trained on 5,000 expert-coded sentences. Sentences may indicate funding instruments, general promotion, restriction, or neutral references. Funding includes subsidies, tax rebates, and special funds. Promotion captures strategic endorsement without a concrete fiscal instrument. Restriction captures export restrictions, capacity-reduction mandates, and environmental compliance orders. Table 10 reports the classification scheme with examples from the original Chinese documents.

Let $\mathcal{A}^+ = \{\text{funding, promotion}\}$ denote supportive policy attitudes. Let $N_{s,t}^{a,N}$ and $N_{s,p,t}^{a,P}$ denote the raw number of national and provincial sentences, respectively, that mention sector s and are classified as attitude a . Let T_t^N and $T_{p,t}^P$ denote total document length in the corresponding national and provincial corpora. The baseline industrial-policy score is the normalized frequency of supportive references: $IP_{s,p,t} = 100 \times \frac{\sum_{a \in \mathcal{A}^+} (N_{s,t}^{a,N} + N_{s,p,t}^{a,P})}{T_t^N + T_{p,t}^P}$. If no provincial document information is available, provincial raw counts and document length are set to zero, so the total score equals the national score.

We then decompose the total score by policy content. For $a \in \mathcal{A}^+$, define $IP_{s,p,t}^a = 100 \times \frac{N_{s,t}^{a,N} + N_{s,p,t}^{a,P}}{T_t^N + T_{p,t}^P}$. By construction, $IP_{s,p,t} = IP_{s,p,t}^{\text{funding}} + IP_{s,p,t}^{\text{promotion}}$. The component-specific

estimates reported below decompose the effect of total policy intensity into concrete fiscal language and broader promotional language. Restrictive signals are normalized analogously but kept separate because they capture capacity controls, export restrictions, and regulatory constraints rather than policy support. The resulting baseline measure varies at the sector-province-year level; national-only measures vary at the sector-year level.

The BERT classifier achieves a mean weighted F1-score of 0.81 (± 0.01) and a mean macro-F1 of 0.76 (± 0.02) across K-fold validation. A dictionary-based approach yields measures with correlation above 0.99 with the BERT-based measures. Appendix C reports validation results.

Interpretation. The index measures documented policy intensity, not realized fiscal expenditure. It captures the share of official policy attention devoted to a sector in a given province-year and attitude category. It does not weight documents by their fiscal magnitude: a planning document announcing a large subsidy program and one announcing a small subsidy program receive the same sentence-level weight. This is appropriate for a measure of documented policy support, but it means that the index should not be interpreted as a monetary measure of transfers. Section 4 validates the index by showing that it predicts realized firm-level subsidy receipts.

The normalization addresses two concerns. First, it accounts for secular growth in document production. Raw document and sentence counts rise over time, but the normalized score captures the share of document content devoted to a sector.² Second, it mitigates heterogeneity in disclosure across provinces. Regions that publish more documents mechanically generate more sector references, but normalization ensures that the relevant variation comes from the composition of policy attention rather than the volume of published text.

Classification uncertainty. The classifier introduces measurement error relative to perfect human coding. Aggregating sentence-level classifications to the sector-province-year level

²Figures 1a and 1b show that the upward trend in raw counts is substantially attenuated after normalization, confirming that part of the increase reflects greater document production rather than intensified policy effort.

reduces idiosyncratic errors. The close agreement between the BERT-based and dictionary-based measures also indicates that the index is not driven by a particular modeling choice. The main remaining concern is vocabulary drift, for example after Made in China 2025. The dictionary-based measure uses fixed keyword lists and produces nearly identical aggregate scores, which mitigates this concern.³

3.2 Other data

Chinese customs data. Transaction-level customs records from the Chinese General Administration of Customs report exports and imports by firm, HS8 product, destination, and year for 2011-2015. We use these data to estimate the response of Chinese export values, destination scope, product scope, and quality-adjusted export prices. These data have been widely used to study Chinese trade (Manova and Zhang, 2012; Brandt et al., 2017; Gong and Xu, 2025).

Chinese NBS firm-level data. The NBS database covers manufacturing firms above the official reporting threshold, equivalent to roughly \$3 million in annual revenue during the sample period. Key variables include employment, gross output, sales, government subsidies, total wages, cost of main business, net fixed assets, depreciation, and input VAT, all reported at the firm-year level with 4-digit Chinese Industrial Classification codes. Intermediate input expenditures are not directly reported for 2011-2014 but can be constructed from cost and wage data, as described in Appendix H. We use the NBS data to estimate production functions, compute firm-level markups, total factor productivity (TFP), and measure domestic responses to industrial policy (Hsieh and Klenow, 2009; Brandt et al., 2017; Barwick et al., 2024).

French customs data. Transaction-level customs data from the French customs administration (DGDDI) record firm-product-destination export and import flows. We use pre-period exports to construct fixed product-destination weights and use 2011-2020 outcomes

³As an additional robustness check, the stability of the relationship between our measure and firm-level subsidies across sub-periods (not reported) confirms that vocabulary drift does not drive the main results.

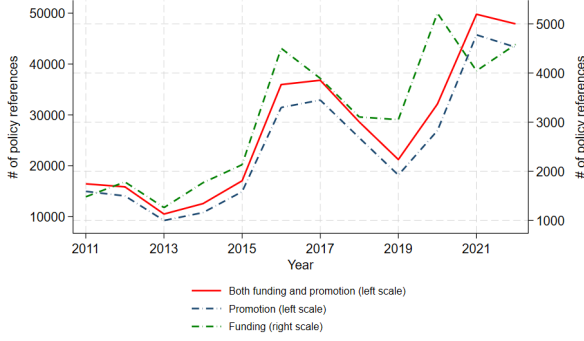
to estimate the effect of exposure to Chinese industrial policy on French exports (Eaton et al., 2011; Mayer et al., 2014).

French firm-level data. The FICUS/FARE balance-sheet dataset provides information on French firms’ production, value added, employment, wage bill, and intermediate-input expenditures for 2011-2020. We use it to estimate how exposure to Chinese industrial policy affects French firms’ domestic operations.

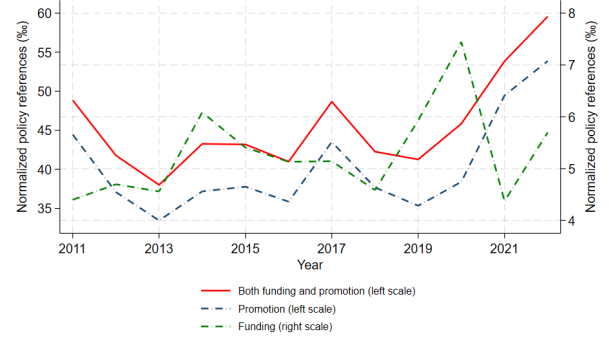
4 Key empirical facts

This section establishes three facts about the policy measure. First, documented industrial policy is widespread, but its intensity varies sharply over time, across administrative levels, and across sectors. Second, the text-based index predicts realized fiscal support. Third, the onset of provincial targeting is followed by an increase in firm-level subsidies. These facts justify using sector-province-year variation in documented policy intensity as the treatment variable in the empirical analysis.

Fact 1: Industrial policy intensity is pervasive, uneven, and rising. Figure 1 reports the evolution of policy references by instrument type from 2011 to 2022. Raw references rise sharply after 2015, the year of Made in China 2025. The increase is broad-based but largest for references that combine financial support with promotional language. A natural concern is that the trend reflects greater document production or disclosure rather than a change in policy priorities. Figure 1b addresses this concern by normalizing sectoral references by total document length. The post-2015 increase remains visible in the normalized series, indicating that industrial sectors account for a larger share of policy attention, not only a larger volume of published text.



(a) Raw counts



(b) Normalized

Figure 1: Evolution of policy references by instrument type

The increase is not only national. Figure 2 shows that provincial governments account for most documented policy activity after normalization. This pattern matters for identification because national policies target many sectors early in the sample, while provincial policies generate staggered sector-province-year variation.

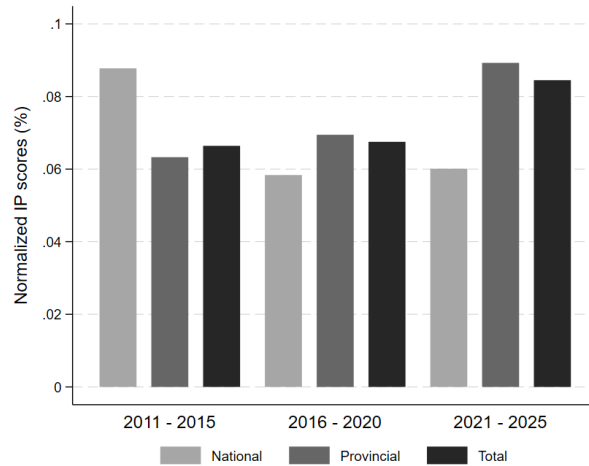


Figure 2: Normalized industrial policy intensity by government level and period

Policy attention is also concentrated across sectors. Figure 3 shows that pharmaceuticals, transport equipment, and specialized machinery dominate manufacturing references, while agriculture absorbs more than one-third of total sector references. Together, these patterns show that the measure contains the cross-sector, cross-province, and over-time variation needed for the empirical design.

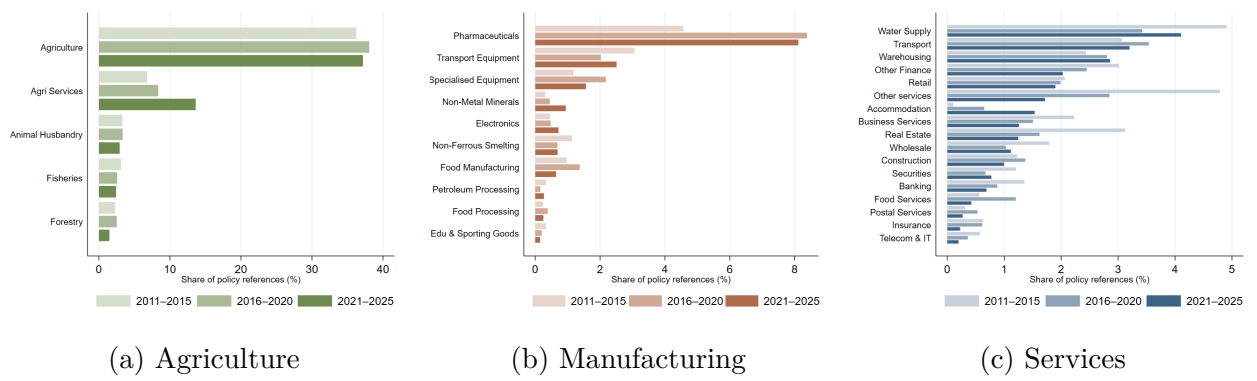


Figure 3: Share of policy references by sector and period

Appendix E documents the spatial counterpart of this variation. Appendix Figure 8 shows that cumulative documented policy intensity is uneven across provinces, and Appendix Figures 9–11 show that this geography differs across broad sectors. These maps are descriptive, but they clarify the source of the empirical variation: policy exposure is not a national sector-year series assigned uniformly across provinces. The identifying comparisons below come from changes in sector-province-year exposure within the regression designs, not from cross-sectional differences across provinces.

Fact 2: Policy intensity predicts realized fiscal support. If policy language reflects real effort rather than rhetoric, sectors receiving greater textual attention should also receive larger fiscal transfers. We aggregate firm-level subsidy accounts from the NBS to the sector-province-year level and regress them on the policy intensity index using PPML with province $\times$ year fixed effects. Table 1 reports a positive and precisely estimated relationship. The coefficient on the funding component is an order of magnitude larger than the coefficient on the promotion component, consistent with the classification scheme: funding language captures concrete fiscal commitments, while promotion captures broader strategic endorsement.⁴ The index should therefore not be interpreted as a monetary measure of transfers: one additional policy reference does not map to a fixed dollar amount of fiscal support. The

⁴The PPML pseudo R^2 is reported only as a descriptive fit statistic. The validation test is whether the policy score has incremental predictive power conditional on province $\times$ year fixed effects, which absorb province-year differences in fiscal capacity and disclosure intensity.

conditional correlation nevertheless shows that the text-based index captures systematic variation in realized policy effort.

Table 1: Correlation between policy intensity and aggregated subsidies

	Aggregated production subsidies		
	(1)	(2)	(3)
IP score (Total)	3.050*** (0.533)		
Funding (Total)		23.708*** (3.510)	
Promoting (Total)			3.299*** (0.596)
Observations	3,579	3,579	3,579
Pseudo R^2	0.270	0.264	0.269
Province $\times$ year FE	Yes	Yes	Yes

Notes: PPML regressions. Dependent variable: production subsidies aggregated at the sector-province-year level. Policy indices are normalized by total document length in each province-year. Standard errors clustered at the sector-province level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Fact 3: Subsidies rise at policy onset. To move beyond contemporaneous correlations, we exploit the staggered timing of provincial targeting in an event-study design. We focus on provincial policy because national policy targets nearly all sectors early in the sample and therefore provides little untreated comparison variation. Provincial policies are staggered across sectors and regions. We estimate:

$$Y_{i,t} = \alpha_i + \gamma_t + \sum_{g \in \mathcal{G}} \sum_{\ell \in \mathcal{L}, \ell \neq -1} \theta_{g,\ell} \mathbf{1}\{G_{s(i),p(i)} = g\} \mathbf{1}\{t - g = \ell\} + \epsilon_{i,t}, \quad (1)$$

where $Y_{i,t}$ is the total amount of subsidies received by firm i , located in province p operating in sector s at the year t , $G_{s,p}$ is the first year in which province p targets industry s , $\mathcal{G}$ is the set of treated cohorts, and $\mathcal{L}$ is the set of observed relative years. The omitted period is the year before targeting, $\ell = -1$. Never-treated sector-province cells form the comparison group. The specification includes firm and year fixed effects; standard errors are clustered

at the sector-province level. Pre-treatment coefficients $\theta_{g,\ell}$ for $\ell < 0$ test for differential trends; post-treatment coefficients for $\ell \geq 0$ estimate cohort-specific treatment effects. The reported coefficient for event time ℓ is $\hat{\tau}_\ell = \sum_{g \in \mathcal{G}_\ell} a_{g,\ell} \hat{\theta}_{g,\ell}$, where $\mathcal{G}_\ell$ contains the treated cohorts observed at relative time ℓ and the aggregation weights $a_{g,\ell}$ sum to one. This is the interaction-weighted event-study approach of [Sun and Abraham \(2021\)](#).

Figure 4 reports the dynamic treatment effects for production subsidies at the firm-level. The pre-period coefficients are close to zero. After policy onset, subsidies rise by 20 to 40 percent relative to the pre-period. Provincial governments do not merely signal support in documents; they also reallocate fiscal resources toward targeted firms.

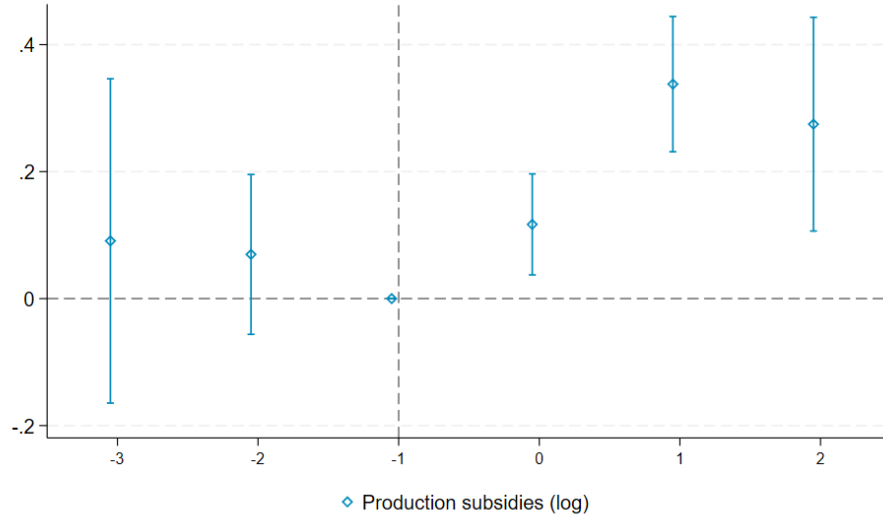


Figure 4: Dynamic effect of provincial industrial policy on production subsidies

Notes: Fully interacted cohort-by-event-time DiD estimator following [Sun and Abraham \(2021\)](#). Firms located in never-treated sector-province cells form the comparison group. Standard errors clustered at the sector-province level.

These facts establish that the text-based measure captures meaningful variation in policy effort. The next section develops a framework that maps this policy variation into firm-level responses, export prices, and foreign displacement.

5 Theoretical framework

We use the linear-demand model of [Melitz and Ottaviano \(2008\)](#) to organize the empirical margins. The benchmark environment has heterogeneous firms, endogenous markups, and selection through a destination-specific cost cutoff. This keeps the markup mechanism close to variable-markup trade models in which market access and competition change residual demand and price-cost margins ([Atkeson and Burstein, 2008](#)). We apply this environment sector by sector. In a treated Chinese sector, industrial policy affects continuing exporters on the intensive margin, affects destination participation on the extensive margin, and changes the competitive environment faced by foreign firms in the same market.

The framework serves three purposes. First, it separates the direct cost effect of industrial policy from the equilibrium response of markups. Second, it distinguishes variable-cost support from fixed-cost market-access support. Third, it delivers the shift-share structure used to measure foreign exposure. Appendix [F](#) contains the derivations. The main text keeps the objects needed for the empirical analysis.

5.1 Industrial policy in the model

This subsection defines the economic environment and the policy object. The primitive demand system is the one-sector Melitz-Ottaviano environment. Sector notation enters only because policy varies across sectors in the data. A sectoral policy shock can shift two primitives. Variable-cost support lowers the delivered marginal costs of Chinese firms. Fixed-cost support lowers the cost of serving a destination. The first margin changes prices, markups, and sales for continuing firms. The second margin changes market access and destination scope.

We follow the standard Melitz-Ottaviano assumptions on preferences, technology, and Pareto heterogeneity.

Preferences and market structure. Destination n has L_n identical consumers. Let x_{in} denote per-capita consumption of variety i , and let $q_{in} = L_n x_{in}$ denote aggregate sales.

Preferences over differentiated varieties in a sector are quasi-linear:

$$u_n = q_{0n} + \alpha \int_{\Omega_n} x_{in} di - \frac{\gamma}{2} \int_{\Omega_n} x_{in}^2 di - \frac{\beta}{2} \left(\int_{\Omega_n} x_{in} di \right)^2, \quad (2)$$

where γ governs own-variety substitutability and β governs cross-variety substitutability. Inverse demand is $p_{in} = \alpha - \gamma q_{in}/L_n - \beta Q_n/L_n$, where $Q_n = \int_{\Omega_n} q_{in} di$. As in Melitz-Ottaviano, the maximum price that yields positive demand is a cost cutoff; markups vary with a firm's distance from this cutoff.

Quasi-linearity eliminates income effects on differentiated-good demand, which renders lump-sum financing of policy support neutral for differentiated-good choices.⁵

Technology and costs. Firms are indexed by origin $o \in \{C, F, \dots\}$, where C is China and F is France. Firm i from o has productivity φ_{io} and wage cost w_o . Serving destination n requires iceberg trade cost $\tau_{on} \geq 1$ and a per-period market-access cost $f_{on} \geq 0$. Delivered marginal cost is

$$c_{in} = \frac{w_o \tau_{on}}{\varphi_{io}}.$$

The fixed cost is paid only by active firms. It captures certification, distribution, compliance, and other destination-specific costs that affect participation without mechanically changing the marginal cost of a continuing exporter.

Entry and productivity distribution. Prospective entrants pay a sunk entry cost f_E denominated in the numeraire, common across origins and sectors, and draw productivity φ from a Pareto distribution with shape $k \geq 1$ and lower support φ_M . Equivalently, delivered costs in destination n have distribution

$$G_{on}(c) = \left(\frac{c}{c_{\max,on}} \right)^k, \quad g_{on}(c) = \frac{k c^{k-1}}{c_{\max,on}^k}, \quad c \in [0, c_{\max,on}],$$

⁵A lump-sum tax used to finance support changes only numeraire consumption q_{0n} . It does not enter differentiated-good demand, pricing, or cutoff conditions.

where $c_{\max,on} \equiv w_o \tau_{on} / \varphi_M$. The upper truncation implies that, conditional on the cutoff c_{Dn} , the distribution of active firms is the original cost distribution truncated at the relevant participation threshold.

Policy shock. Industrial policy is origin-specific and sector-specific. In a targeted Chinese sector s , variable-cost support reduces Chinese delivered marginal costs proportionally:

$$c_{in}(\psi_v) = \frac{w_C(\psi_v) \tau_{Cn}}{\varphi_{iC}^0} \exp(-\psi_v). \quad (3)$$

Equivalently, in a treated sector, one can write $\log \varphi_{iC,s} = \log \varphi_{iC,s}^0 + \psi_v$ holding wages and trade costs fixed. The marginal-cost formulation makes clear that the primitive policy channel is a cost reduction. For a continuing exporter in a treated sector,

$$\frac{d \log c_{in}}{d \psi_v} = \frac{d \log w_C}{d \psi_v} - 1.$$

Fixed-cost support lowers the cost of serving a destination:

$$f_{Cn}(\psi_f) = f_{Cn} e^{-\rho \psi_f}, \quad \rho \geq 0, \quad (4)$$

where f_{Cn} denotes the pre-policy fixed cost. This margin operates on the destination extensive margin. It changes whether a firm can profitably serve a market, but it does not directly change the delivered marginal cost of an active incumbent: $d \log c_{in} / d \psi_f = 0$.

The variable-cost formulation also covers standard marginal-cost subsidies: an ad valorem subsidy that changes marginal cost from c_i to $c_i(1 - \sigma)$ is locally equivalent to a log-productivity gain $\psi_v \simeq \sigma$.⁶

⁶The equivalence follows from $\log[c_i(1 - \sigma)] - \log c_i = \log(1 - \sigma) \simeq -\sigma$. Fixed-cost support is different because it changes participation without directly changing the marginal cost of a continuing exporter.

5.2 Equilibrium

We work within a sector s and suppress the sector index where unambiguous. The key equilibrium object is the destination cost cutoff c_{Dn} . As in Melitz-Ottaviano, this cutoff summarizes market toughness: a lower c_{Dn} means that only lower-cost firms sell, residual demand is more elastic, and markups are lower.

Intensive margin. Consider a firm that sells in destination n . It takes c_{Dn} as given and chooses price to maximize variable profit. The first-order condition gives:

$$p_{in} = \frac{c_{Dn} + c_{in}}{2}, \quad \mu_{in} \equiv \frac{p_{in} - c_{in}}{p_{in}} = \frac{c_{Dn} - c_{in}}{c_{Dn} + c_{in}}. \quad (5)$$

Conditional on selling, all firm-level outcomes are functions of the firm's delivered cost and the destination cutoff. Lower-cost firms set lower prices, sell more, and charge higher markups. A fall in c_{Dn} is pro-competitive: it lowers prices and compresses markups for all active firms. Under CES demand, this channel is absent because markups are fixed by the elasticity of substitution. Appendix F.1 derives demand, optimal pricing, revenues, and variable profits.

Substituting (5) into the demand system:

$$q_{in} = \frac{L_n}{2\gamma}(c_{Dn} - c_{in}), \quad (6)$$

$$r_{in} = \frac{L_n}{4\gamma}(c_{Dn}^2 - c_{in}^2), \quad (7)$$

$$\pi_{in} = \frac{L_n}{4\gamma}(c_{Dn} - c_{in})^2. \quad (8)$$

Extensive margin. A firm serves destination n only if variable profit covers the market-access cost f_{on} . Define

$$h_{on} \equiv \sqrt{\frac{4\gamma f_{on}}{L_n}}, \quad \tilde{c}_{Dn}^o \equiv c_{Dn} - h_{on}. \quad (9)$$

Origin- o firms serve destination n iff $c_{in} \leq \tilde{c}_{Dn}^o$. The term $\tilde{c}_{Dn}^o$ is the destination-specific participation cutoff. When $f_{on} = 0$, $h_{on} = 0$ and the participation cutoff coincides with the Melitz-Ottaviano cost cutoff c_{Dn} .

Free entry. Free entry pins down the cost cutoffs. Expected net profit from destination n for a potential entrant from origin o is:

$$\mathbb{E}[\pi_{on} - f_{on}]^+ = \frac{L_n}{2\gamma(k+1)(k+2)} \frac{(c_{Dn} - h_{on})^{k+1} [c_{Dn} + (k+1)h_{on}]}{c_{\max,on}^k}. \quad (10)$$

The free-entry condition sums these expected net profits over destinations:

$$\sum_n L_n (c_{Dn} - h_{on})^{k+1} [c_{Dn} + (k+1)h_{on}] \tau_{on}^{-k} = B w_o^k, \quad (11)$$

where $B \equiv 2\gamma(k+1)(k+2)f_E/\varphi_M^k$ is common across origins and sectors. After the policy shock, the Chinese free-entry condition in the treated sector becomes:

$$\sum_n L_n [c_{Dn} - h_{Cn} e^{-\rho\psi_f/2}]^{k+1} [c_{Dn} + (k+1)h_{Cn} e^{-\rho\psi_f/2}] \tau_{Cn}^{-k} e^{k\psi_v} = B w_C^k. \quad (12)$$

The comparative statics below are local around interior equilibria with $0 \leq h_{on} < c_{Dn} < c_{\max,on}$ for the relevant origin-destination pairs, positive entry, and a nonsingular free-entry Jacobian. These restrictions ensure positive participation thresholds and locally well-defined cutoff responses. Appendix F.2 derives the participation threshold, expected net profit, and free-entry conditions. The important implication for the empirical analysis is that policy affects continuing exporters directly through c_{in} and indirectly through the equilibrium cutoff c_{Dn} .

5.3 Comparative statics of industrial policy

The comparative statics follow from the two margins above. Variable-cost support shifts the delivered cost of Chinese firms. Both variable-cost and fixed-cost support can also shift the

destination cutoff through entry and selection. Let

$$\Delta \equiv 1 - \frac{d \log w_C}{d \psi_v} > 0. \quad (13)$$

Thus Δ is the log reduction in Chinese delivered cost from variable-cost support, net of the Chinese wage response. The equilibrium cost cutoff moves according to:

$$\frac{d \log c_{Dn}}{d \psi_v} = -\varepsilon_{Dn}^v \Delta, \quad \frac{d \log c_{Dn}}{d \psi_f} = -\varepsilon_{Dn}^f. \quad (14)$$

The elasticities ε_{Dn}^v and ε_{Dn}^f summarize the strength of cutoff compression in destination n . Positive values mean that Chinese policy makes competition tougher in that market. Appendix F.3 derives these elasticities from free entry. Appendix F.4 derives the comparative statics below. The propositions consider exposed destinations with $\varepsilon_{Dn}^v \geq 0$ and $\varepsilon_{Dn}^f \geq 0$.

Proposition 1 (Chinese intensive margin: prices and markups). *Suppose $\Delta > 0$ and consider a continuing Chinese exporter with $c_{in} < \tilde{c}_{Dn}^C$ in an exposed destination n . Then*

$$\frac{d \log p_{in}}{d \psi_v} = -\Delta \frac{c_{Dn} \varepsilon_{Dn}^v + c_{in}}{c_{Dn} + c_{in}} < 0, \quad (15)$$

$$\frac{d \log p_{in}}{d \psi_f} = -\frac{c_{Dn}}{c_{Dn} + c_{in}} \varepsilon_{Dn}^f, \quad (16)$$

$$\frac{d \mu_{in}}{d \psi_v} = \frac{2c_{in}c_{Dn}}{(c_{Dn} + c_{in})^2} \Delta (1 - \varepsilon_{Dn}^v), \quad (17)$$

$$\frac{d \mu_{in}}{d \psi_f} = -\frac{2c_{in}c_{Dn}}{(c_{Dn} + c_{in})^2} \varepsilon_{Dn}^f, \quad (18)$$

The sign of $d \mu_{in}/d \psi_v$ is the sign of $1 - \varepsilon_{Dn}^v$, and $d \mu_{in}/d \psi_f \leq 0$ when $\varepsilon_{Dn}^f \geq 0$.

The intensive-margin prediction is not simply that subsidies lower prices. Variable-cost support lowers the firm's own delivered cost, which tends to raise the price-cost margin, but it can also lower the market cutoff, which reduces residual demand and compresses markups. When cutoff compression is strong, prices fall while markups move little. This is the empirical pattern tested with quality-adjusted export prices and firm-level markups.

Proposition 2 (Chinese sales and destination margin). *Suppose $\Delta > 0$ and consider a continuing Chinese exporter with $c_{in} < \tilde{c}_{Dn}^C$ in an exposed destination n . Then*

$$\frac{d \log r_{in}}{d\psi_v} = \frac{2\Delta(c_{in}^2 - \varepsilon_{Dn}^v c_{Dn}^2)}{c_{Dn}^2 - c_{in}^2}, \quad (19)$$

$$\frac{d \log r_{in}}{d\psi_f} = -\frac{2\varepsilon_{Dn}^f c_{Dn}^2}{c_{Dn}^2 - c_{in}^2}. \quad (20)$$

The destination extensive margin satisfies:

$$\frac{d \log(c_{in}/\tilde{c}_{Dn}^C)}{d\psi_v} = -\Delta + \frac{c_{Dn}\varepsilon_{Dn}^v \Delta}{\tilde{c}_{Dn}^C}, \quad (21)$$

$$\frac{d\tilde{c}_{Dn}^C}{d\psi_f} = \frac{\rho h_{Cn}}{2} - c_{Dn}\varepsilon_{Dn}^f. \quad (22)$$

Variable-cost support raises incumbent export revenue iff $c_{in}/c_{Dn} > \sqrt{\varepsilon_{Dn}^v}$. Destination scope expands under ψ_v iff $\varepsilon_{Dn}^v < \tilde{c}_{Dn}^C/c_{Dn}$, and under ψ_f iff $\rho h_{Cn}/2 > c_{Dn}\varepsilon_{Dn}^f$.

The sales and destination-margin results show why export values and destination counts need not move mechanically with prices. A continuing exporter can charge a lower price without increasing revenue if the cutoff falls enough to reduce residual demand. Fixed-cost support works through market access: it expands destination scope only when the direct reduction in the participation cost dominates the pro-competitive fall in the cutoff. Appendix F.5 reports the corresponding aggregate-revenue expressions.

5.4 Effects on French firms

Chinese policy affects foreign firms only through the destination cutoff. A French firm does not receive the policy shock, so its delivered cost is fixed. When Chinese policy lowers c_{Dn} in a product-destination market, the French firm's residual demand shifts inward. Appendix F.6 derives the revenue and employment expressions below.

Proposition 3 (French intensive-margin displacement). *Consider a French firm active in*

an exposed destination n , with $c_{iF} < c_{Dn}$. Then

$$\frac{d \log r_{iF,n}}{d\psi_v} = -\nu_{iF,n} \varepsilon_{Dn}^v \Delta, \quad \frac{d \log r_{iF,n}}{d\psi_f} = -\nu_{iF,n} \varepsilon_{Dn}^f, \quad \nu_{iF,n} \equiv \frac{2c_{Dn}^2}{c_{Dn}^2 - c_{iF}^2} > 1. \quad (23)$$

The displacement elasticity $\nu_{iF,n}$ is increasing in c_{iF}/c_{Dn} . French firms closer to the cutoff lose more revenue, in percentage terms, when Chinese policy tightens competition in their market.

Aggregating across the firm's destination portfolio gives:

$$d \log r_i^F = - \sum_n \omega_{in}^F \nu_{iF,n} \left(\Delta \varepsilon_{Dn}^v d\psi_v + \varepsilon_{Dn}^f d\psi_f \right), \quad (24)$$

where $\omega_{in}^F = r_{iF,n}/R_{iF}$. This expression is the theoretical counterpart of the empirical exposure measure. A French firm is more exposed when its pre-period revenue is concentrated in markets where Chinese policy generates stronger cutoff compression.

Proposition 4 (French employment). *Under a competitive French labor market with fixed wage w_F , employment of French firm i in market n is:*

$$\ell_{iF,n} = \frac{L_n c_{iF} (c_{Dn} - c_{iF})}{2\gamma w_F}. \quad (25)$$

Since c_{iF} is fixed, employment falls for all active French firms in markets where Chinese policy lowers c_{Dn} .

5.5 Input-output linkages

The baseline model treats sectors independently. We add an accounting layer for input-output propagation because industrial policy in one sector can change costs in linked sectors. Let $\alpha_{ss'}$ denote sector s 's cost share of inputs from sector s' . Let $\pi_s^v < 0$ and $\pi_s^f \leq 0$ denote the average price response in sector $\hat{s}$ to variable-cost and fixed-cost support. Let $\lambda_s > 0$ summarize the pass-through from upstream demand expansion into sector- s input

prices, and let $\eta \in [0, 1]$ allow fixed-cost support to generate weaker demand pressure than variable-cost support. Appendix F.7 gives the accounting behind the propagation equation.

Proposition 5 (Input-output propagation). *The total change in log effective cost for Chinese sector- s firms attributable to the vector of industrial policies is:*

$$\Delta \log c_s^{\text{IO}} = -\Delta_s \psi_{v,s} + \sum_{\hat{s}} \alpha_{s\hat{s}} \left(\pi_{\hat{s}}^v \psi_{v,\hat{s}} + \pi_{\hat{s}}^f \psi_{f,\hat{s}} \right) + \lambda_s \sum_{\hat{s}} \alpha_{\hat{s}s} (\psi_{v,\hat{s}} + \eta \psi_{f,\hat{s}}), \quad (26)$$

where the first term is the own-sector cost reduction, the second term is the downstream input-cost channel, and the third term is the upstream demand channel.

The input-output extension clarifies the sign ambiguity of indirect effects. Policy in sector $\hat{s}$ lowers costs for downstream users of $\hat{s}$, but it can raise costs for upstream suppliers if the expansion of $\hat{s}$ increases demand for their inputs. At the firm-product-destination level, the direct variable-cost channel predicts lower quality-adjusted export prices for continuing exporters, while the indirect input-output channel has an ambiguous sign.

6 Impact of industrial policies on Chinese firms

This section tests the Chinese-firm implications of Propositions 1 and 2. The central margin is export performance: export sales and destination scope measure the intensive and participation margins in Proposition 2. We report coefficients and the implied change from zero policy exposure to the 90th and 99th percentiles of $IP_{s,p,t}$ in the relevant estimation sample. Quality-adjusted prices and markups discipline the cost channel in Proposition 1; additional Chinese-firm outcomes that are not primary model margins are reported in Appendix G.⁷

The empirical object is the total policy score $IP_{s,p,t}$ defined in Section 3. The regressions use the text-based index as a reduced-form measure of policy intensity, not as a measure of realized transfers. This choice is deliberate. Realized subsidies are chosen jointly with firm outcomes, while the text index captures fiscal and non-fiscal instruments. Table 1 shows

⁷These outcomes are useful for interpreting firm adjustment, but they are not the primary margins delivered by the model.

that the index is related to subsidy receipts, but the empirical specifications below estimate the total effect of documented policy exposure.

A separate concern is relocation across provinces. Firm fixed effects absorb time-invariant location and productivity differences. Within-firm variation comes from changes in policy intensity in the firm’s sector-province cell. Because the NBS identifier tracks establishments, relocating firms appear as exits and entries rather than as within-firm changes.

6.1 Export sales and destination scope

Proposition 2 predicts that Chinese policy can raise export sales for continuing exporters when the direct cost reduction dominates the pro-competitive cutoff response. It also predicts expansion in destination scope when the policy-induced reduction in market-access costs dominates cutoff compression. We map these margins into firm-year customs outcomes: export value measures aggregate export sales, and the number of destinations measures market participation.

For export outcome $Y_{i,t}$, we estimate the PPML specification:

$$Y_{i,t} = \exp \left\{ \beta IP_{s(i),p(i),t} + \alpha_i + \delta_{p(i),t} \right\} \times u_{i,t}, \quad (27)$$

where α_i are firm fixed effects, $\delta_{p,t}$ are province-by-year fixed effects, and $u_{i,t}$ is the multiplicative PPML error. Companion columns replace province-by-year fixed effects with year fixed effects. Standard errors are clustered at the sector-province level.

Table 2 reports the export-margin estimates. The preferred specifications are columns (2) and (4), which absorb province-year shocks. The total IP coefficient in column (2) is 0.171. In the customs sample, no exposure corresponds to $IP_{s,p,t} = 0$ and the 90th percentile is 0.0583. Moving from no exposure to the top decile of policy exposure raises firm export value by 1.0 percent, relative to a baseline mean of about \$1.5 million.⁸ Since Chinese

⁸Monetary baseline means are expressed in U.S. dollars using average bilateral exchange rates over the relevant customs sample. These conversions affect only descriptive means; coefficient estimates are unchanged. Percentage effects for log-link and log-outcome specifications are computed as $100[\exp(\hat{\beta}\Delta IP) - 1]$.

industrial policies are concentrated, it is relevant to focus on the impact on the top 1%. Moving from no exposure to the top percentile increases firm export value by 6.5%, which is substantial. The same move raises the number of destinations by 0.44 percent, relative to a mean of 9.0 destinations. The export expansion therefore operates through sales and destination reach. Product-scope estimates are reported in Appendix G. They are useful for distinguishing destination expansion from product expansion, but product scope is not the primary market-access margin in line with Proposition 2.

Table 2: Industrial policy and export performance

	Export value		# destinations	
	(1)	(2)	(3)	(4)
Panel A. IP score	0.209*** (0.050)	0.171*** (0.047)	0.053** (0.023)	0.076*** (0.026)
Panel B. Funding score	1.100*** (0.295)	0.870** (0.403)	0.334** (0.160)	0.415** (0.202)
Panel C. Promotion score	0.234*** (0.060)	0.192*** (0.053)	0.058** (0.025)	0.086*** (0.029)
Observations	1,160,871	1,160,871	1,160,871	1,160,871
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No
Province $\times$ year FE	No	Yes	No	Yes

Notes: PPML regressions. Each panel reports a separate regression. Standard errors are clustered at the sector-province level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The component estimates are descriptive decompositions of the total index. Funding has larger per-unit coefficients than promotion, but funding exposure is much more concentrated in the data. The baseline quantitative statement therefore uses the total normalized score.

The policy-onset design provides the dynamic counterpart to Table 2. It is not a substitute for the continuous-intensity estimates. It asks whether outcomes move when documented targeting begins and whether treated sector-province cells show differential pre-trends. Figure 5 reports event-study estimates for export value and export participation in the NBS panel using Equation 1. The pre-onset coefficients do not show a systematic divergence of treated sector-province cells before targeting. After onset, both outcomes move in

the direction predicted by Proposition 2. The figure therefore supports the timing of the export response, while Table 2 provides the main quantitative magnitudes.

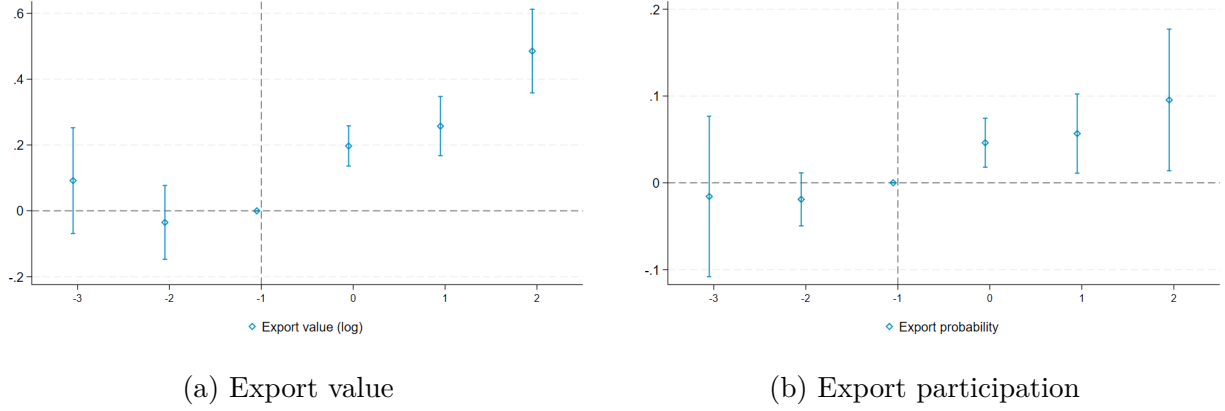


Figure 5: Dynamic effects of provincial industrial policy on export outcomes
Notes: Fully interacted cohort-by-event-time DiD estimator following [Sun and Abraham \(2021\)](#). Never-treated sector-province cells form the comparison group. Standard errors are clustered at the sector-province level.

6.2 Quality-adjusted export prices

Proposition 1 predicts that variable-cost support lowers prices for continuing Chinese exporters. To separate prices from quality, we construct quality-adjusted unit values following [Khandelwal et al. \(2013\)](#); [Fan et al. \(2015\)](#); [Manova and Yu \(2017\)](#). For firm i , product h , destination d , and year t , quality is recovered from:

$$\sigma \ln p_{i,h,d,t} + \ln q_{i,h,d,t} = \lambda_{i,d,t} + \epsilon_{i,h,d,t}, \quad (28)$$

where $p_{i,h,d,t}$ is the unit value, $q_{i,h,d,t}$ is quantity, $\lambda_{i,d,t}$ is a firm-destination-year fixed effect, and $\sigma = 5$. The quality-adjusted price is $\ln \bar{p}_{i,h,d,t} = \ln p_{i,h,d,t} - \hat{\epsilon}_{i,h,d,t}$. $\lambda_{i,d,t}$ collects the potential fixed cost faced by the firm to reach country d , but also the destination price index and the market income.

Table 3 reports the price results for the pooled sample and for the Broad Economic Categories most directly connected to the export-competition mechanism: intermediate and

final goods.⁹ The pooled coefficient is small and not statistically different from zero. The final-goods coefficient is -0.267 in the preferred specification. With the 90th percentile of total policy exposure equal to 0.0563 in the price sample, moving from no exposure to the top decile lowers quality-adjusted final-good prices by 1.5 percent. Although the decrease may seem small, it is quite substantial when moving to the top percentile, with a 16% drop. Intermediate-goods prices do not respond precisely. The remaining Broad Economic Categories are kept in Appendix G. They serve as diagnostics for heterogeneity across product classes rather than as separate model tests.

Table 3: Industrial policy and quality-adjusted export prices

	Log quality-adjusted unit value		
	All goods (1)	Intermediate goods (2)	Final goods (3)
Panel A. IP score	-0.044 (0.031)	-0.043 (0.058)	-0.267^{***} (0.062)
Panel B. Funding score	-0.632^{**} (0.290)	-0.633 (0.511)	-1.583^{***} (0.520)
Panel C. Promotion score	-0.042 (0.033)	-0.039 (0.064)	-0.300^{***} (0.070)
Observations	862,330	673,237	532,455
Firm FE	Yes	Yes	Yes
Province $\times$ year FE	Yes	Yes	Yes

Notes: OLS regressions. Each panel reports a separate regression. Quality adjustment follows Khandelwal et al. (2013). Standard errors are clustered at the sector-province level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Input-output linkages provide an additional propagation margin. Proposition 5 shows that the sign of indirect price effects is ambiguous because policy in linked sectors can lower input costs and can also raise demand for upstream inputs. For this reason, the input-output estimates are not part of the primary export-margin test. Appendix G.1 reports the all-goods specification for quality-adjusted export prices. The coefficient on IO-weighted policy

⁹We classified products using the last revision *BEC Rev. 5* into four categories: primary, capital, intermediate, and final goods. For final and intermediate goods, we removed primary goods and only kept processed products. We removed the agriculture and government sectors due to their specificity.

exposure is -0.049 , with a standard error of 0.025 , indicating that linked-sector exposure is associated with lower quality-adjusted export prices on average. We treat this result as a diagnostic for propagation across Chinese sectors, not as a separate displacement margin.

6.3 Firm-level markups

The markup prediction is weaker than the price prediction. Proposition 1 shows that variable-cost support raises destination-specific markups only if the direct cost reduction dominates cutoff compression, while fixed-cost support lowers markups through the cutoff. The empirical markup is a firm-level object, constructed from production data using the [De Loecker and Warzynski \(2012\)](#) approach, and averages across domestic and export markets. We therefore treat markups as an additional implication of the export-price response rather than as a separate primitive mechanism. Appendix H gives the production-function estimation and markup construction.

Appendix H provides two diagnostics that ask whether the markup result masks heterogeneity that would weaken the export-price mechanism. The question is not whether policy exposure lowers markups for all firms. Proposition 1 allows variable-cost support to raise markups when the direct cost reduction dominates the induced change in the cutoff. The relevant prediction for the empirical mechanism is narrower: if lower quality-adjusted export prices reflect pass-through of cost or productivity improvements, exporters should not display a broad positive markup response to policy exposure. Table 18 shows that the total policy score is positively correlated with markups in the full sample and among domestic-oriented firms, but the exporter coefficients are smaller and not statistically different from zero. Funding language has larger positive correlations, again concentrated among domestic-oriented firms. Table 19 interacts policy exposure with the firm’s export share. The interaction is negative for all policy measures and statistically significant for funding, implying that the markup association weakens as firms become more export oriented. These patterns are aligned with the model prediction taken to the data: documented policy exposure lowers export prices, and among exporters this price response is not accompanied by a systematic

increase in firm-level markups.

The policy-onset design is the preferred timing test for markups. Figure 6 reports event-study estimates for log firm-level markups around provincial policy onset. The coefficients are centered near zero before and after onset for all firms, domestic-oriented firms, and exporters. This is the markup counterpart of the price result: policy exposure lowers quality-adjusted export prices, but does not generate detectable increases in average firm-level markups. The appendix diagnostics are therefore useful for interpretation, but the main evidence on timing comes from the event study.

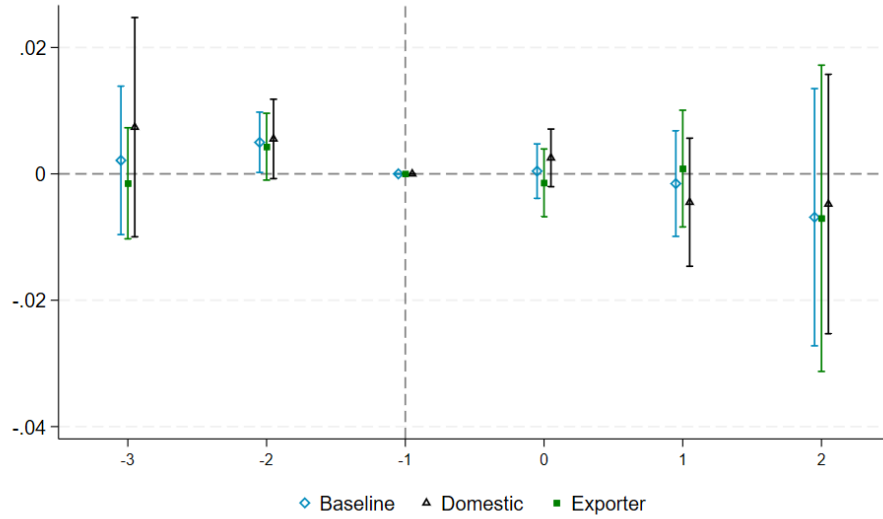


Figure 6: Dynamic effect of provincial industrial policy on firm-level markups

Notes: Event-study coefficients from estimator (1), with log De Loecker-Warzynski markup as the outcome. Markers report estimates for all firms, domestic-oriented firms, and exporters. Vertical bars are 95% confidence intervals. Standard errors are clustered at the sector-province level.

The Chinese results deliver the margins used in the spillover test: policy exposure raises export sales and destination reach, lowers quality-adjusted final-good prices, and leaves firm-level markups unchanged in the policy-onset design. Section 7 uses these margins to test the foreign-firm displacement implication of Proposition 3.

7 International spillovers: evidence from French firms

This section tests the foreign-firm implications of Propositions 3 and 4. The central margin is French export revenue in product markets exposed to Chinese industrial policy. We interpret the preferred coefficient as the change in French firm-product exports when exposure moves from zero to the 90th percentile of the product-level policy distribution. Firm-level production and employment are reported as downstream outcomes of export displacement. Wage-bill and import responses are reported in the appendix because they are not the primary displacement margins of the model. The input-output price-propagation diagnostics belong with the Chinese price results, since they concern propagation across Chinese sectors rather than French displacement.

7.1 French export revenue

Proposition 3 predicts that a foreign firm’s revenue falls when Chinese policy lowers the destination cutoff in markets where the foreign firm is active. The empirical counterpart is French firm-product export revenue. We assign each HS6 product to a Chinese industry using the NBS concordance and merge the product with the Chinese industrial-policy score. This avoids a many-to-many sector concordance between Chinese industries and French firm industries.

For French firm i , product h , and year t , we estimate:

$$X_{i,h,t} = \exp \{ \beta IP_{h,t} + \alpha_{i,t} + \gamma_h \} \times v_{i,h,t}, \quad (29)$$

where $X_{i,h,t}$ is export value, $\alpha_{i,t}$ are firm-by-year fixed effects, γ_h are product fixed effects, and $v_{i,h,t}$ is the multiplicative PPML error. Firm-by-year fixed effects absorb all firm-level shocks in a year. Identification comes from comparing products with different Chinese policy exposure within the same French firm-year. Standard errors are clustered at the Chinese industry level.

Table 4 reports the estimates. The preferred total-index coefficient is -0.413 in Panel B. In the French export sample, zero exposure corresponds to $IP_{h,t} = 0$ and the 90th percentile of total policy exposure is 0.0616. Moving from no exposure to the top decile of Chinese policy exposure reduces French firm-product exports by 2.5 percent, relative to a baseline mean of about \$470,000. However, the most exposed firms (top percentile) experienced a decrease by about 25%. The national-document estimates in Panel A are also negative. Component estimates are descriptive: promotion loads closely on the total index, while funding is precisely estimated only for national documents.

Table 4: Chinese industrial policy and French firm-product exports

	Export value		
	IP score (1)	Funding score (2)	Promotion score (3)
Panel A. National	-0.292^{***} (0.087)	-1.522^{***} (0.323)	-0.325^{***} (0.099)
Panel B. Total	-0.413^{**} (0.190)	-1.143 (0.763)	-0.471^{**} (0.217)
Observations	9,085,859	9,085,859	9,085,859
Firm $\times$ year FE	Yes	Yes	Yes
Product FE	Yes	Yes	Yes

Notes: PPML regressions. The outcome is French firm-product export value. Each column in each panel reports a separate regression. Standard errors are clustered at the Chinese industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

7.2 Firm-level displacement

Aggregating (23) across a firm's export portfolio gives the shift-share expression in (24). The empirical exposure measure weights Chinese policy by the firm's pre-period export portfolio and by China's baseline importance in the product-destination market:

$$\text{Exposure}_{i,t}^X = \sum_d \sum_h z_{d,h,t_0}^{CN} IP_{h,t} w_{i,d,h,t_0}^X, \quad (30)$$

where w_{i,d,h,t_0}^X is firm i 's pre-period export share in product-destination pair (h, d) and z_{d,h,t_0}^{CN} is China's baseline import share in that market. Baseline weights use $t_0 = 2004$, before the estimation period.

We estimate:

$$Y_{i,t} = \beta \text{Exposure}_{i,t}^X + \alpha_i + \gamma_t + \epsilon_{i,t}, \quad (31)$$

where $Y_{i,t}$ is log production or log employment. Firm fixed effects absorb time-invariant firm and industry characteristics. Production is the firm-level counterpart of revenue displacement, and employment maps to Proposition 4. Wage-bill effects and French import responses are reported in Appendix I.

Table 5 reports the firm-level estimates. The total export-exposure coefficient is negative for production and close to zero for employment. Funding exposure generates the sharpest firm-level displacement: the coefficients are -1.154 for production and -0.954 for employment. These estimates indicate that the export-revenue losses in Table 4 are reflected in firms' domestic scale, especially when the Chinese policy shock is fiscal.

Table 5: Chinese industrial policy exposure and French firm outcomes

	Dependent variable (log)	
	Production (1)	Employment (2)
Export exposure: IP score	-0.187^{**} (0.074)	-0.034 (0.064)
Export exposure: funding	-1.154^{***} (0.373)	-0.954^{***} (0.363)
Export exposure: promotion	-0.202^{**} (0.087)	-0.016 (0.075)
Observations	427,686	428,165
R^2	0.963	0.965
Firm FE	Yes	Yes
Year FE	Yes	Yes

Notes: OLS regressions on French exporting firms. Exposure is defined in (30). Standard errors are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Input-output linkages in Proposition 5 concern propagation across Chinese sectors, not

the French displacement margin. We therefore keep them outside the main spillover test. Appendix G.1 reports the corresponding price-propagation estimates; they are useful diagnostics but are not used to construct the French exposure measure.

8 Conclusion

This paper provides a sector-province-year measure of Chinese industrial policy and uses it to study how policy support is transmitted through trade. The measure is based on official government documents, validated against firm-level subsidy receipts, and matched to Chinese and French firm data. The empirical design exploits staggered provincial targeting across sectors; the theory clarifies the margins that should move when policy lowers marginal costs or reduces destination fixed costs.

The evidence points to a price-competition mechanism. Targeted Chinese firms receive more subsidies, expand exports, serve more destinations, and lower quality-adjusted export prices. Firm-level markups do not rise in event-study estimates, which indicates that the average response is not rent capture through higher price-cost margins. The cost reduction instead appears to pass through to export prices.

The same mechanism generates international spillovers. French firms lose export revenue in products more exposed to Chinese industrial policy, and the losses are larger in markets where China is an important competitor. Firm-level production also falls. The paper therefore connects a large, decentralized system of industrial policy to measurable changes in Chinese competitiveness and to displacement of foreign firms in international product markets.

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Appendices

A Data methodology

This appendix documents the construction of the text-based industrial-policy measures used in Section 3. It records the data source, sampling restrictions, classification procedure, validation exercises, and normalization of the raw text measures. The central object is the sector-province-year score $IP_{s,p,t}$, defined as the sum of normalized funding and promotion components. Funding and promotion components describe the composition of this support. Restrictive references are measured separately.

A.1 Reading China Better and alternative sources

The empirical measure relies on the *Reading China Better* (RCB) database (Brussee, 2025). RCB collects documents directly from official Chinese government and Party websites, including central ministries, provincial governments, selected local authorities, and state-owned media. The database was designed for text-as-data analysis of Chinese policy and stores both document metadata and original Chinese text. The version used for this project contains national, ministerial, and provincial documents issued between 2011 and 2024.

RCB is suited to the present research design for three reasons. First, it preserves the original Chinese text rather than relying on translated summaries. This is important because our classifier uses sentence-level language to identify policy attitudes toward sectors. Second, it covers multiple levels of government, which is necessary in a system where provincial governments implement central priorities. Third, the database records document type and issuing date, allowing us to restrict the sample to document categories with more reliable disclosure.

The main alternative source for cross-country industrial policy is the *Global Trade Alert*. GTA is valuable for international comparability, but it provides English-language summaries prepared by editors rather than the full original documents. This creates a measurement issue for our purpose: a classifier applied to summaries may recover editorial language rather than policy language. Beida Fabao provides broad coverage of Chinese legal and policy documents and is useful for institutional mapping. Our approach makes a different choice. We prioritize a narrower corpus with explicit controls for disclosure heterogeneity, original full text, and supervised validation, because the empirical design requires a treatment measure with known statistical properties.

A.2 Sampling frame and dates

The sampling frame follows the institutional logic in Section 2. We retain documents issued by national authorities, ministries, and provincial People’s Governments. Provincial documents are central because they capture implementation. We do not systematically include lower-level departments in the baseline measure. Local departments often disclose documents inconsistently, and in the Chinese administrative hierarchy they usually implement directives rather than define major industrial-policy priorities independently.

We restrict the period to 2011–2024. The start date corresponds to the 12th Five-Year Plan and follows the stabilization of disclosure after the Open Government Information framework. RCB distinguishes the issuing date of a document from its public release date.

When the two differ, we use the issuing date, because this is the relevant date for the policy signal faced by firms.

We also restrict document types. The baseline sample includes *opinions* (*yijian*, 意见) and planning documents: *guihua* (规划), *fang'an* (方案), *jihua* (计划), and *gangyao* (纲要). These documents are consistently published and contain the strategic directives through which industrial policy is communicated. After applying these restrictions and retaining full-text documents, the baseline corpus contains 69,388 documents.

A.3 Construction of the policy score

Let $\mathcal{A}^+ = \{\text{funding, promotion}\}$ denote supportive policy attitudes. Let $N_{s,t}^{a,N}$ and $N_{s,p,t}^{a,P}$ denote the raw number of national and provincial sentences, respectively, that mention sector s and are classified as attitude a . Let T_t^N and $T_{p,t}^P$ denote total document length in the corresponding national and provincial corpora. The baseline industrial-policy score is

$$IP_{s,p,t} = 100 \times \frac{\sum_{a \in \mathcal{A}^+} (N_{s,t}^{a,N} + N_{s,p,t}^{a,P})}{T_t^N + T_{p,t}^P}.$$

The factor 100 expresses the score as supportive references per 100 sentences. If no provincial document information is available, provincial raw counts and document length are set to zero, so the total score equals the national score.

We then decompose the total score by policy content. For $a \in \mathcal{A}^+$, define

$$IP_{s,p,t}^a = 100 \times \frac{N_{s,t}^{a,N} + N_{s,p,t}^{a,P}}{T_t^N + T_{p,t}^P}.$$

By construction, $IP_{s,p,t} = IP_{s,p,t}^{\text{funding}} + IP_{s,p,t}^{\text{promotion}}$. We report funding and promotion separately in the main tables because they have different substantive content. Funding captures concrete fiscal instruments. Promotion captures strategic endorsement without a specific fiscal commitment. Restrictive signals are normalized analogously but kept separate because they may operate through capacity controls, export restrictions, or regulatory constraints rather than through support.

Normalizing by total document length addresses two sources of measurement error. First, document production rises over time. Raw counts would partly reflect the volume of published text rather than the composition of policy attention. Second, transparency differs across provinces. Normalization ensures that a province publishing more documents does not mechanically receive a higher policy score unless a larger share of its policy language is devoted to the sector.

B Transparency patterns

The central data-quality concern is non-random disclosure. Chinese authorities do not publish all policy documents, and disclosure varies across years, regions, and document types. This section documents why the baseline sample starts in 2011, why it focuses on opinions and planning documents, and why lower-level departmental documents are excluded from the baseline.

Provincial disclosure. Table 6 reports disclosure rates for selected provincial and prefecture-level organs in the highest-authority document series, labelled *fa* (发). Early years show large and irregular differences across regions. Disclosure is more stable after 2010, which motivates the 2011 start date.

Table 6: Document disclosure rates for selected provincial and prefecture-level organs (%)

Year	Beijing	Zhejiang	Shandong	Ningxia	Xiamen	Wenzhou
2000	90.7	6.0	–	0.9	5.3	8.7
2001	86.7	15.4	–	1.8	14.0	3.8
2002	91.9	6.2	–	2.5	63.1	8.3
2003	88.6	79.0	–	56.2	72.3	82.8
2004	84.2	69.4	–	49.1	71.1	70.0
2005	41.2	65.1	–	44.7	63.4	70.4
2006	76.0	37.7	–	41.7	67.5	40.1
2007	64.1	34.6	12.7	48.9	64.1	48.2
2008	48.8	25.6	93.7	41.8	59.5	75.3
2009	36.7	52.8	94.0	38.3	75.2	73.0
2010	46.8	38.2	86.9	40.3	72.8	69.6
2011	81.3	44.5	–	24.1	73.5	46.7
2012	77.6	49.6	–	20.7	75.6	67.3
2013	88.6	67.0	77.5	58.5	57.6	62.4
2014	82.4	83.6	90.5	39.0	61.4	55.9
2015	88.1	89.7	95.0	72.8	77.6	55.7
2016	82.8	95.3	98.5	65.3	68.6	83.9
2017	92.1	93.2	99.1	71.9	78.8	76.5
2018	91.8	89.6	96.9	82.8	70.7	69.2
2019	96.4	91.4	100.0	89.7	57.9	67.5
2020	88.7	47.3	100.0	94.7	71.4	74.7
2021	97.8	80.3	100.0	93.1	65.8	63.8
2022	96.4	70.0	99.7	92.2	67.1	71.1
2023	97.2	66.3	100.0	88.0	73.6	59.3
2024	94.7	60.0	99.5	84.9	43.1	50.6

Notes: Replicated from [Brussee \(2025\)](#) using Reading China Better data. The midrule separates the pre-2011 period from the analysis period. Lower-authority document series generally have lower and more volatile disclosure rates.

Disclosure also differs by document type. In the early 2000s, publicly available documents were disproportionately regulatory. By the 2020s, the disclosed corpus contained many more planning and opinion documents. Because document types use different language and perform different administrative functions, shifts in disclosure composition can generate spurious shifts in measured policy attitudes. The baseline sample therefore holds document type fixed.

Department-level disclosure. Table 7 shows that disclosure by local departments is more volatile than disclosure by People’s Governments. This pattern supports the baseline decision to focus on national, ministerial, and provincial People’s Government documents.

Table 7: Document disclosure rates for selected local departments (%)

Year	Lishui Construction	Zibo Construction	Donggang Dev. & Reform	Chengdu Civil Affairs	Zhejiang Health
2000	–	–	–	–	0.4
2005	0.6	–	–	10.1	1.3
2010	1.6	–	–	45.0	2.3
2011	0.4	1.6	–	15.4	5.8
2015	13.1	–	18.5	17.5	21.8
2016	60.8	–	59.9	2.3	42.4
2017	19.8	8.9	37.4	31.0	73.0
2018	88.6	48.7	3.4	19.2	59.1
2019	92.0	51.9	4.5	12.5	52.9
2020	92.9	90.7	2.7	35.6	38.4
2021	90.7	82.6	27.9	48.2	29.8
2022	84.5	39.9	62.8	33.3	26.8
2023	74.4	50.3	67.3	20.5	10.0
2024	81.7	64.6	75.9	13.0	11.3

Notes: Replicated from [Brussee \(2025\)](#). Missing values indicate that disclosure rates cannot be reliably inferred for that source-year. The volatility reinforces the choice to exclude local departments from the baseline corpus.

C Classification details

This section documents the construction and validation of the sector and policy-attitude classifiers.

Sector identification. We identify sector references using a dictionary based on the China Securities Regulatory Commission industry classification and the National Bureau of Statistics industrial classification ([National Bureau of Statistics of China, 2017](#)). Two analysts compiled and reviewed the dictionary: one native Chinese speaker trained in law and politics, and one specialist in Chinese governance and text-as-data methods. The procedure yields 83 sector categories.

Table 8: Classification dictionary for industrial sectors

Industry code	Keywords
A01	农业，耕地红线，粮食安全，农田保护，农田建设，退耕还林还草
A03	苗木，造林，林业科技，森林采伐，林长制，经济林
A05	畜牧业，养殖污染，养殖技术，畜禽
A07	渔业，水产养殖，休渔，网箱
A09	农业机械化，土地经营权流转，乡村振兴，集体林权制度改革，国有林场改革，森林保险，育种，种苗培育，农机，良种繁殖
B01	煤炭采选，煤炭开采
B03	石油开采，天然气开采，页岩开采，页岩气，油气资源

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Table 8 continued

Industry code	Keywords
B05	锰矿开采, 铬矿开采, 铁矿开采, 钢铁原料矿
B07	铜矿开采, 铝矿开采, 铅锌矿开采, 锡矿开采, 镍矿开采, 有色金属矿, 贵金属矿采选, 稀土矿开采
B09	砂石开采, 砂石资源, 采砂, 非金属矿采选
B49	其他采矿业, 地热, 天然水
B50	勘探, 地面沉降, 矿坑回填
C01	农副食品加工业, 农产品初加工, 农产品地理标志
C03	食品生产许可, 食品加工, 追溯体系
C05	饮料制造, 饮料饮品制造, 特色植物饮料
C11	纺织业, 纺织废水, 纺织制造业, 印染行业, 废旧纺织品, 纺织品出口
C13	服装制造, 服装生产, 服装产业, 服装外贸
C14	皮革制造, 毛皮加工, 羽绒制品
C21	木材加工, 竹业, 藤制品, 棕制品, 草制品
C25	家具制造, 传统家具
C31	造纸行业, 造纸产业, 纸制品产业
C35	印刷工业, 数字印刷
C37	体育用品, 儿童用品, 学生用品, 体育器材标准, 文教体育用品行业
C41	焦化, 炼化一体化, 石化化工行业, 炼油行业
C43	化学原料, 化学品制造, 肥料制造, 涂料制造
C47	化纤行业, 化纤新材料, 化纤生产
C48	橡胶制造, 天然橡胶, 再生橡胶
C49	工程塑料, 通用塑料, 新型塑料, 塑料污染
C51	基础电子元器件产业, 电子元器件
C55	智能家居, 小家电, 日用电子设备
C59	电子废弃物, 电子产品维修, 电子设备维修
C61	建材行业, 水泥玻璃行业, 金属矿行业, 绿色建材
C65	钢铁产业, 钢铁工业
C67	有色金属产业, 新能源金属, 合金
C69	金属制品
C71	通用设备制造, 机械制造
C73	专用设备制造, 石油钻采, 半导体专用设备, 医疗器械, 高端制造
C75	交通设备制造, 汽车工业, 新能源汽车, 造车, 造船业, 航空制造业, 船舶制造业, 造船
C76	电机制造, 输配电设备, 光伏设备, 电池行业
C78	工业自动化, 专用仪器, 仪表制造
C81	医药制造, 中医药, 医药研发, 制药行业
C85	疫苗, 生物制品, 血液制品
D03	液化石油气生产, 天然气生产, 煤气生产, 天然气供应, 燃气经营
D05	供水, 污水处理, 海水淡化, 供水官网
E01	建筑业, 建筑工程, 工程施工, 住建业, 土木工程

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Table 8 continued

Industry code	Keywords
E05	建筑装饰装修, 装饰装修, 建筑拆除
F01	铁路旅客运输, 铁路货物运输, 铁路价格, 中欧班列
F03	城市公共交通, 道路货物运输, 治超限载, 农村公路
F05	油气管网, 天然气管道, 管道高后果
F07	海上客运, 游轮, 轮渡, 国际航运
F09	国产民机运营, 航空运输, 通用航空
F11	交通运输辅助, 公路养护, 空管, 母港建设, 火车站建设, 铁路维护
F21	仓储, 冷库, 危化品仓库, 物流枢纽
G81	通信设备制造, 通信设备
G83	计算机制造, 物联网设备, 算力基建, 电子信息制造业
G85	通信服务, 农村宽带, 光纤入户, 网络服务
G87	软件开发, 物联网技术服务, 互联网运维, 遥感, 计算机软件
H01	食品批发, 烟草专卖, 烟草批发, 日用品批发
H03	煤炭经营, 成品油批发, 大宗商品
H09	综合批发, 商贸物流
H11	零售
I01	农村合作社, 商业银行服务, 农村资金互助社, 融资服务, 网络借贷, 银行监管
I11	人身保险, 财产保险, 商业养老, 保险中介, 保险监管, 保险资产管理
I21	资本市场服务, 证券行业, 公募基金, 私募基金, 期货
I31	信托
I99	金融信息, 金融资产管理, 融资担保, 债券担保
J01	房地产开发, 保障性住房, 经济适用房, 公租房
J05	物业管理
J09	房地产中介, 房地产估价, 房屋检验
K01	市政设施, 市容管理, 城市绿化, 景区管理
K10	邮政服务
K20	科研服务业
K30	餐饮服务, 餐饮业, 食品经营
K32	旅馆业, 民宿, 旅游住宿
K34	文旅产业, 旅游, 导游
K36	娱乐场所, 演艺产业, 非遗演出, 娱乐经营, 娱乐服务
K37	养老机构, 社区养老, 社区护理, 保健行业
K39	经营租赁
L01	出版审批, 出版物经营, 出版物市场, 出版物交易, 网络出版
L05	音像制品, 电子出版物, 音乐产业, 电影电视业
L15	非遗艺术, 艺术行业, 艺术品, 公共艺术
L20	新闻信息服务, 融媒体, 新闻业
L99	文化产业, 文化事业, 文创开发

Policy-attitude classifier. We classify the attitude of each sector sentence using a BERT-base Chinese model. Pre-processing removes bracketed text, duplicates, inconsistent spacing and punctuation, and uninformative sentences with fewer than five Chinese characters. Numbers and dates are masked. The classifier uses a linear layer, a ReLU layer, dropout, and a second linear layer. Hyperparameters are fixed before validation: learning rate 2×10^{-5} , batch size 16 for training and 32 for validation, dropout 0.3, warmup share 0.1, and focal-loss parameter 2.

The codebook defines four categories. Funding captures concrete fiscal instruments such as subsidies, tax rebates, special project funds, government guidance funds, special government bonds, and direct government procurement. Promotion captures general support for a sector without a specific fiscal instrument. Restriction captures measures that limit sectoral development, reduce investment or production, or revoke existing support. Neutral captures factual statements, titles, headers, and administrative references without a policy attitude.

Table 9: Examples for policy attitude

Tag	Typical vocabulary
Restriction	严格监管, 限制, 应当/必要, 执行
Promotion	支持, 发展, 推进
Funding	专项资金, 所得税优惠政策, 消费券, 补贴政策, 政府性融资担保, 省现代产业引导基金, 财政投入

Table 10: Overview of sentence classifications

Attitude	Explanation	Example sentences
Funding	Concrete calls to set up funding instruments for the sector, including subsidies, tax rebates, special project funds, and other fiscal benefits.	“Support poverty alleviation; eco-migrants are entitled to grain subsidies, seed subsidies, animal husbandry subsidies, agricultural machine subsidies, ecological compensation, and other policies.” 扶贫生态移民在流转土地承包经营权后, 按规定享受粮食直补、良种补贴、畜牧补贴、农机补贴、生态补偿等各类强农惠农富农政策 (Guangxi Province, 2014) “Provide special financial funds to express delivery enterprises for customs clearance, warehousing, and distribution.” 省级财政专项资金要对快递企业在跨境通关业务、仓配一体化服务、区域总部中心建设 (Hunan Province, 2017)

Continued on next page

Table 10 continued

Attitude	Explanation	Example sentences
Promotion	General support for the development of a sector without a specific fiscal instrument.	“Encourage social capital to invest in the operation of public elderly institutions.” 鼓励引导社会资本投资运营公办养老机构 (Suzhou City, 2015) “Actively support pharmaceutical projects disseminating traditional Chinese medicine abroad.” 积极支持医药类项目在国外传播中医药文化 (Ministry of Commerce, 2022) “Vigorously develop frontier industries such as integrated circuits, artificial intelligence, biomedicine, and intelligent new-energy vehicles.” 大力发展集成电路、人工智能、生物医药、智能新能源汽车 (Shanghai City, 2021)
Restriction	Active discouragement of the sector or calls to reduce investment and production.	“Coke ovens with carbonization chamber height below 5.5 meters and without national coking-access approval shall be included in the elimination category.” 对晋政发〔2005〕13号文件登记的项目变更炉型的焦炉：炭化室高度小于5.5米、未通过国家焦化准入公告的焦化项目办理炉型变更手续后，列入相应类别淘汰年限淘汰 (Shanxi, 2011)

Notes: The classifier also includes a neutral category for sentences that do not express a clear attitude toward the sector.

Document distribution. In total, 55,892 documents in the retained corpus reference one or more identified sectors. Figure 7 reports their distribution by administrative level and Five-Year Plan period. The concentration of documents around the start of Five-Year Plans is consistent with the planning cycle described in Section 2.

Validation. The training and testing dataset consists of 5,000 sentences that mention at least one sector. The sample is stratified by sector to limit the risk that the model learns sector frequency rather than policy attitude. The codebook was developed iteratively. Two coders first labeled 250 sentences and reached 88% agreement. After codebook refinement, they labeled another 250 sentences and reached 95.6% agreement. One coder then completed the remaining labels.

We evaluate the model using five-fold cross-validation. The classifier achieves a mean weighted F1-score of 0.813 (± 0.011) and a mean macro-F1 of 0.761 (± 0.020). Tables 11 and 12 report per-class validation metrics and the mean confusion matrix across folds.

The main classification ambiguity is between funding and promotion. This is expected because fiscal instruments are often embedded in broader promotional language. The confusion between funding and restriction is limited, which matters for separating supportive from restrictive policy signals.

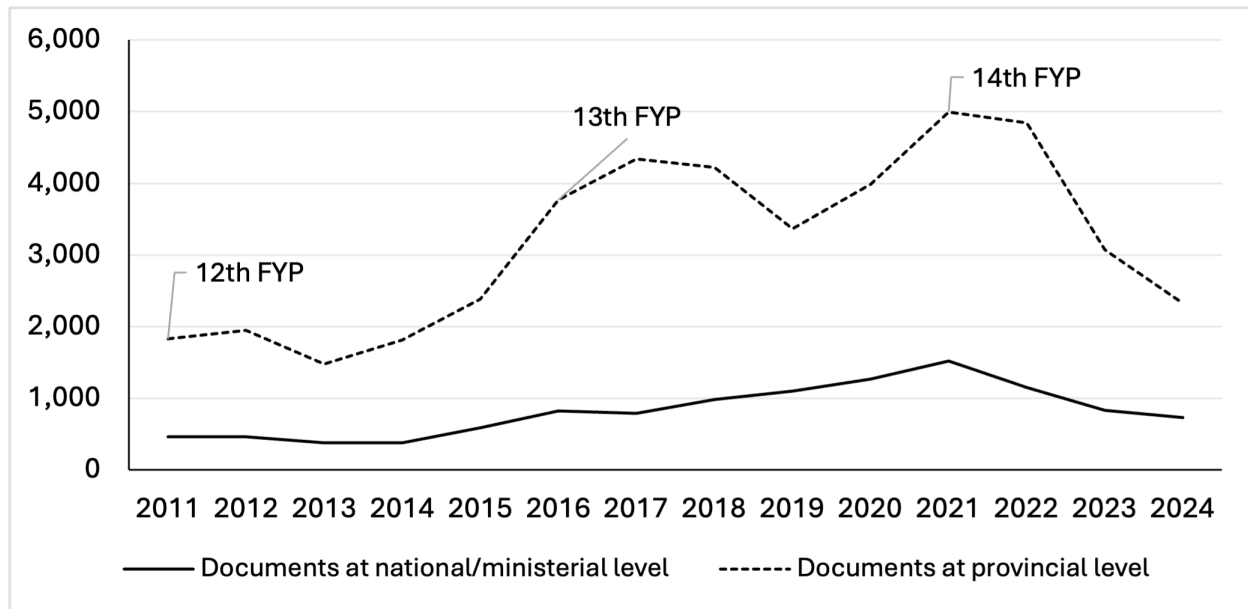


Figure 7: Distribution of sector-referencing documents by administrative level and Five-Year Plan period

Table 11: Validation metrics

	Precision mean	Precision std.	Recall mean	Recall std.	F1 mean	F1 std.
Funding	0.610	0.044	0.830	0.064	0.702	0.046
Neutral	0.756	0.027	0.707	0.022	0.730	0.013
Restriction	0.670	0.032	0.820	0.019	0.737	0.024
Promotion	0.904	0.014	0.847	0.016	0.874	0.008

Table 12: Confusion matrix

	Funding	Neutral	Restriction	Promotion
Funding	33.2	0.8	1.2	4.8
Neutral	4.2	142.2	24.8	30.0
Restriction	3.6	9.4	111.0	11.4
Promotion	13.6	36.0	29.0	435.4

Notes: Entries are averages across validation folds. Rows denote hand-coded classes and columns denote predicted classes.

D Dictionary approach robustness

We construct a dictionary-based measure as a robustness check. Dictionary methods are transparent and replicable, and they perform well in formulaic bureaucratic texts. Chinese policy language is useful in this respect because it has limited morphological variation and relies on stable administrative vocabulary.

Table 13: Classification dictionary for policy attitude

Tag	Keywords
Restriction	限制, 淘汰, 禁止新增, 严禁新增
Promotion	鼓励, 提升, 扶持, 大力发展, 支持, 培育, 积极发展, 加快发展
Funding instruments	政府债券, 专项资金, 免征, 政府投资, 补贴, 财政激励, 补助
Regulation	严格监管, 规范, 健全, 完善

Table 14 reports correlations between the BERT-based and dictionary-based measures. The total IP score has a correlation of 0.995 across methods. The funding and promotion components are also strongly correlated. The close agreement indicates that the empirical measure is not driven by a particular classifier.

Table 14: Correlation between BERT and dictionary measures

	IP score dictionary	IP score BERT	Funding dictionary	Funding BERT	Promotion dictionary	Promotion BERT
IP score dictionary	1.000					
IP score BERT	0.995	1.000				
Funding dictionary	0.829	0.802	1.000			
Funding BERT	0.902	0.893	0.885	1.000		
Promotion dictionary	0.997	0.995	0.784	0.881	1.000	
Promotion BERT	0.992	0.999	0.782	0.869	0.995	1.000

E Spatial distribution of industrial policy

Appendix Figure 8 and Figures 9–11 report the spatial distribution of cumulative provincial policy intensity over 2011–2024. These maps are descriptive. They show that documented industrial policy is geographically uneven and that the relevant variation is not only national or sectoral, but also provincial. This motivates the sector-province-year measurement used in the main text; identification comes from the regression designs, not from cross-sectional map comparisons.

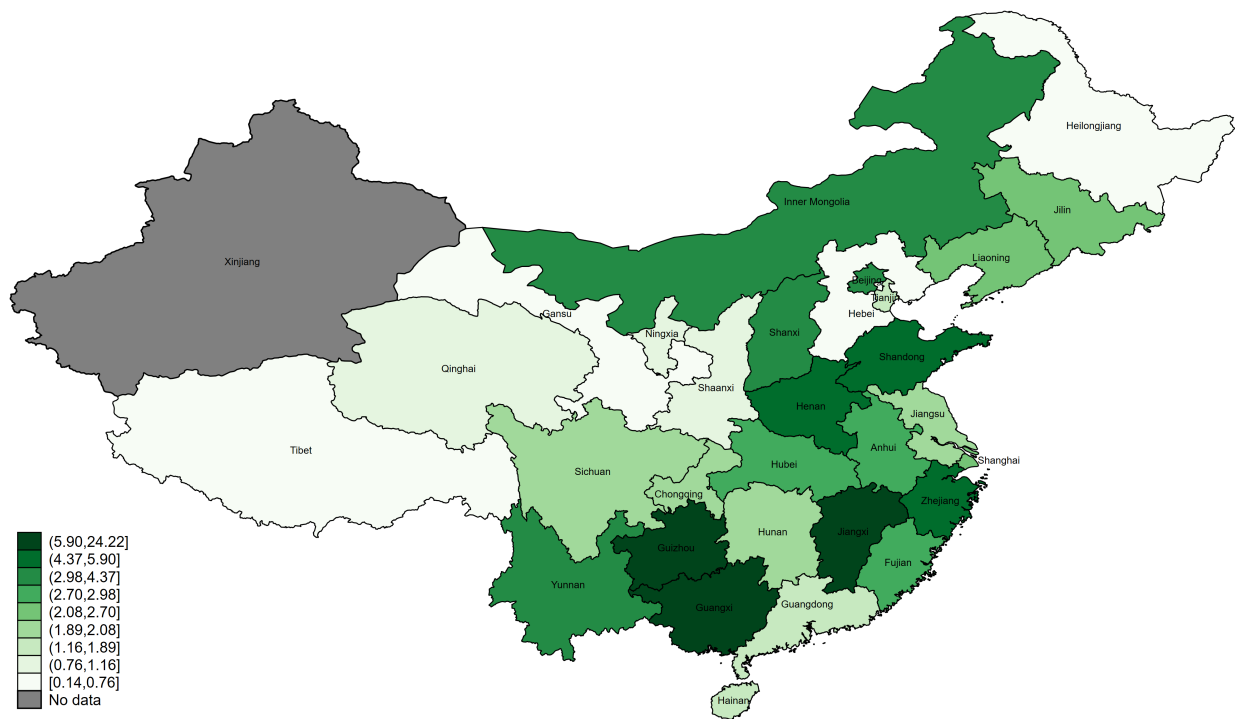
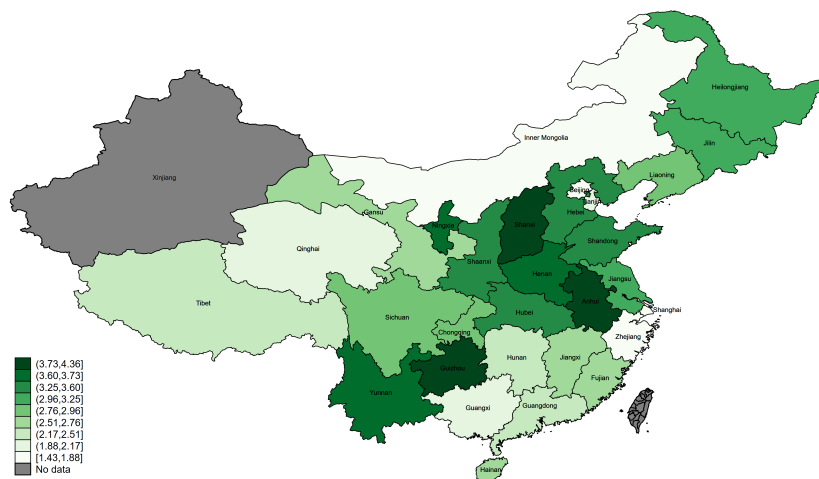
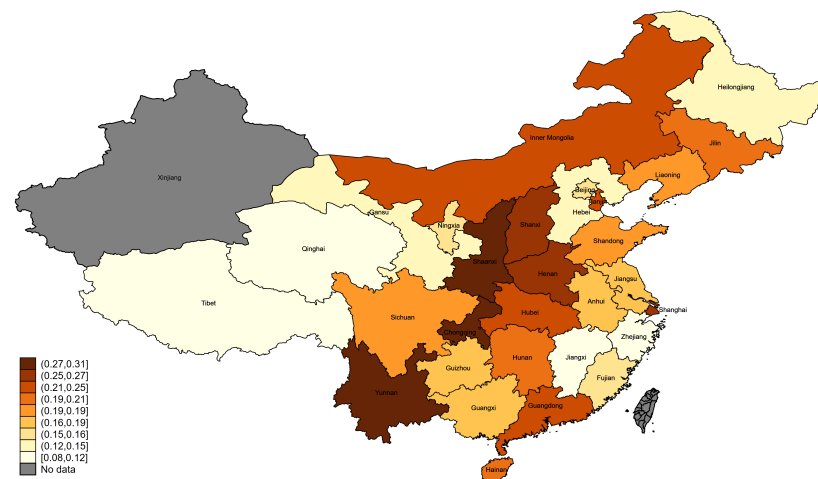


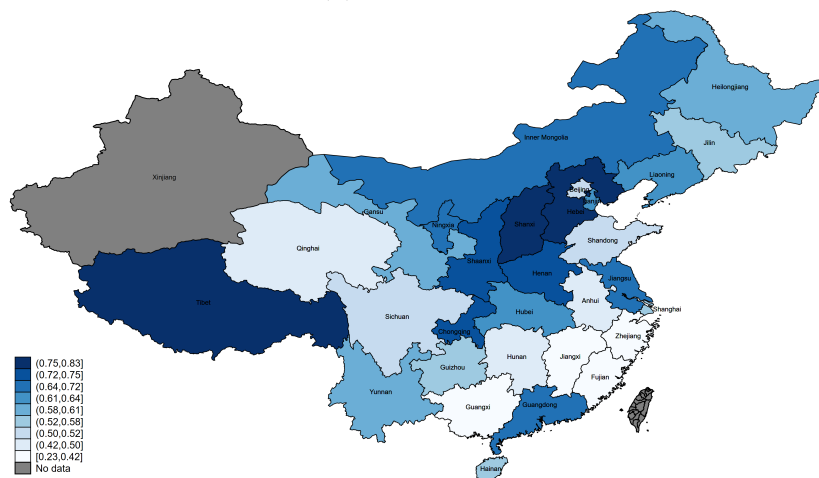
Figure 8: Spatial distribution of cumulative industrial-policy intensity
Notes: The figure reports cumulative funding and promotion references by province from 2011 to 2024, normalized by the national total in provincial documents. No data are available for Xinjiang.



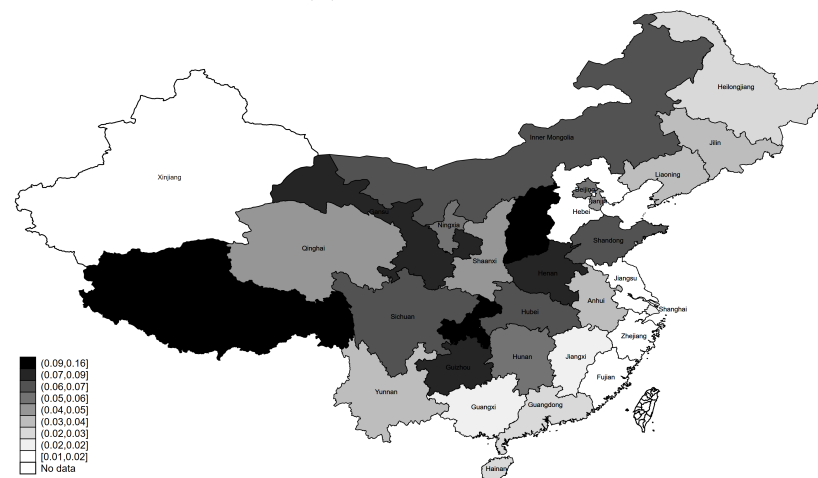
(a) Agriculture



(b) Manufacturing



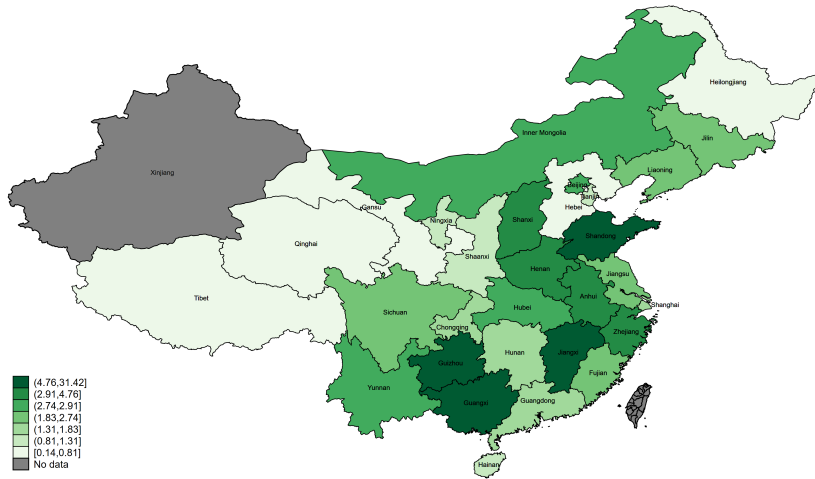
(c) Services



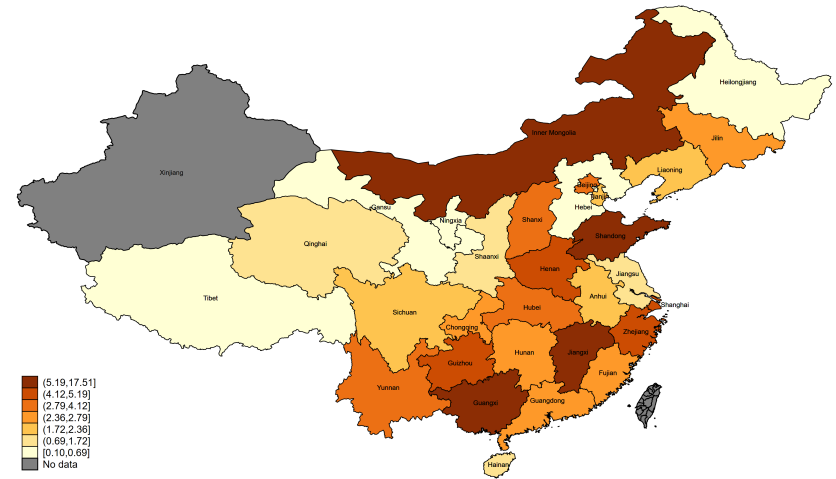
(d) Mining

Figure 9: Spatial distribution of industrial-policy intensity by broad sector

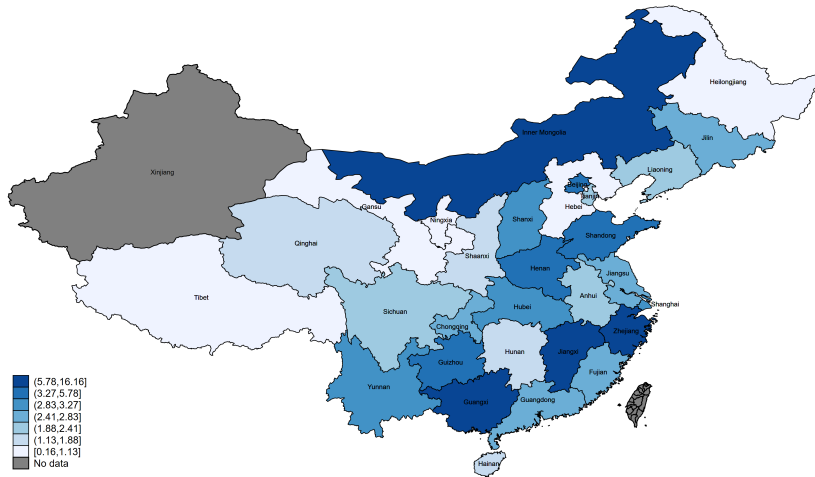
Notes: Intensity is measured as cumulative sector-province references divided by total sentences in the province. No data are available for Xinjiang.



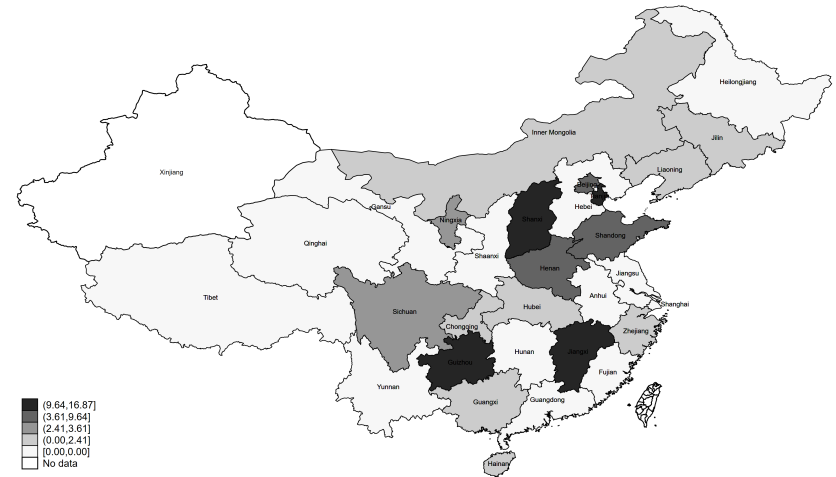
(a) Agriculture



(b) Manufacturing



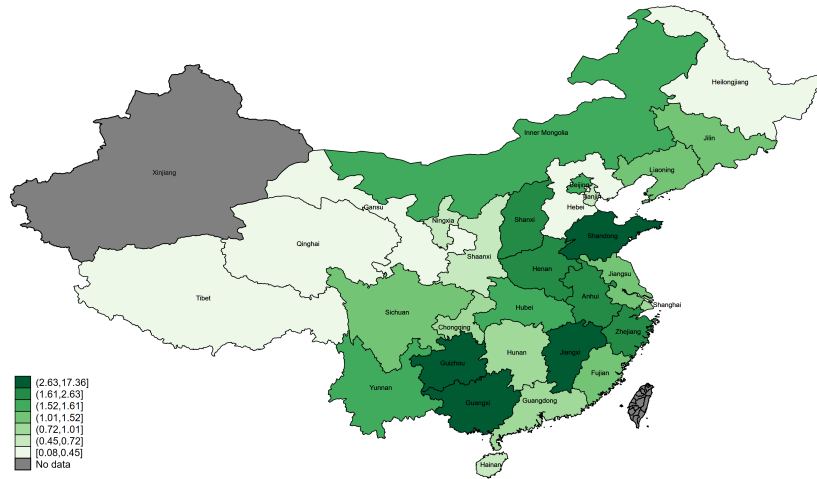
(c) Services



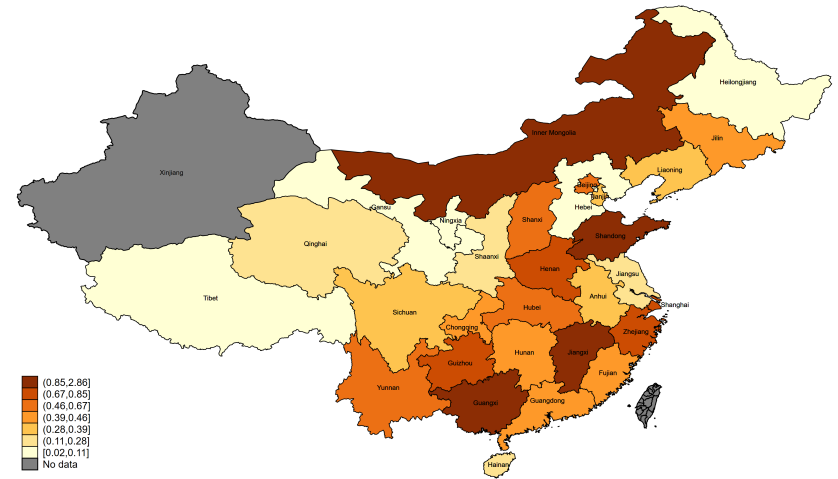
(d) Mining

Figure 10: Provincial concentration within broad sectors

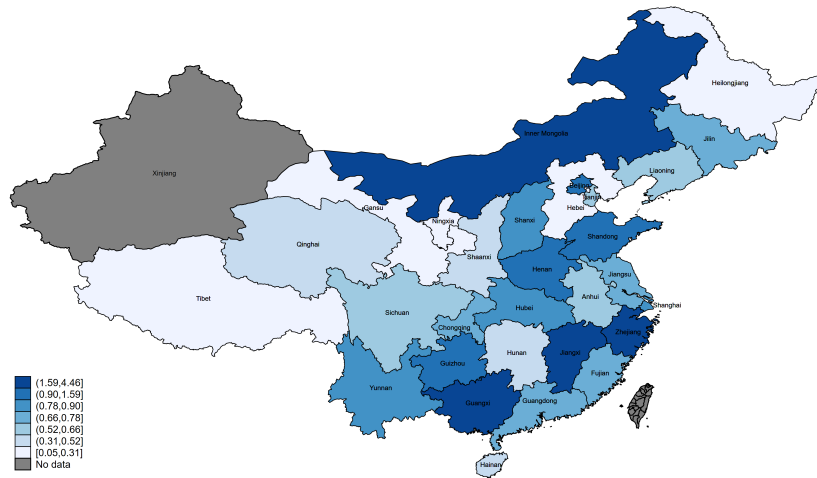
Notes: Intensity is measured as cumulative sector-province references divided by total references in the broad sector.



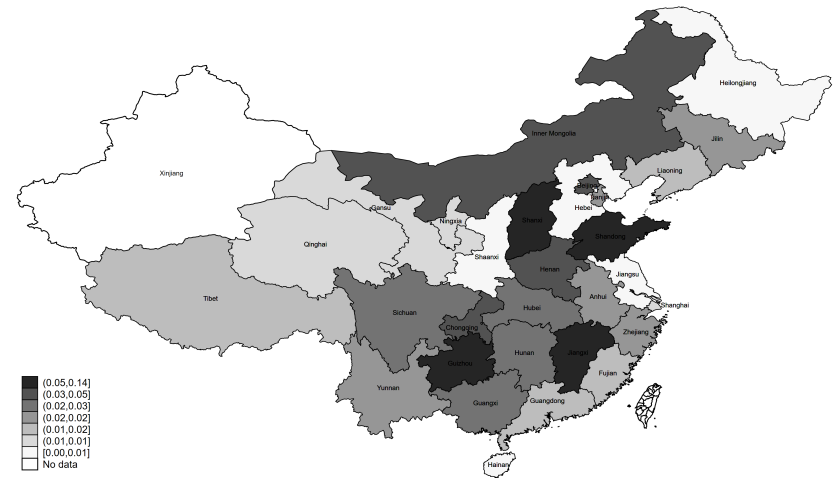
(a) Agriculture



(b) Manufacturing



(c) Services



(d) Mining

Figure 11: Sector-province shares in national policy intensity

Notes: Intensity is measured as cumulative sector-province references divided by total references across all sector-province pairs.

F Model derivations

F.1 Preferences, demand, and optimal pricing

Maximizing (2) subject to the budget constraint yields the FOC for each active variety:

$$p_{in} = \alpha - \gamma q_{in}/L_n - \beta Q_n/L_n.$$

A firm with $c_{in} = c_{Dn}$ sells zero, so $p_{Dn} = \alpha - \beta Q_n/L_n \equiv c_{Dn}$. Substitution gives

$$p_{in} = c_{Dn} - \gamma q_{in}/L_n, \quad q_{in} = \frac{L_n(c_{Dn} - p_{in})}{\gamma}.$$

Profit maximization over p_{in} yields $p_{in} = (c_{Dn} + c_{in})/2$, which implies (6)–(8).

F.2 Fixed costs and free entry

Assume $0 \leq h_{on} < c_{Dn} < c_{\max, on}$ for each origin-destination pair. A firm from origin o is active in destination n iff variable profit covers the fixed cost:

$$\frac{L_n}{4\gamma}(c_{Dn} - c_{in})^2 \geq f_{on}.$$

This gives the participation threshold in (9). Delivered cost has density $g_{on}(c) = kc^{k-1}/c_{\max, on}^k$. Expected net profit from destination n is

$$\mathbb{E}[\pi_{on} - f_{on}]^+ = \frac{L_n k}{4\gamma c_{\max, on}^k} \int_0^{c_{Dn} - h_{on}} [(c_{Dn} - c)^2 - h_{on}^2] c^{k-1} dc.$$

The integral is

$$\int_0^{c_D - h} [(c_D - c)^2 - h^2] c^{k-1} dc = (c_D - h)^{k+1} \frac{2[c_D + (k+1)h]}{k(k+1)(k+2)}.$$

Substitution gives (10) and the free-entry conditions (11)–(12). When $h = 0$, expected profit reduces to

$$\mathbb{E}[\pi_{on}] = \frac{L_n}{2\gamma(k+1)(k+2)} \frac{c_{Dn}^{k+2}}{c_{\max, on}^k},$$

the standard Melitz-Ottaviano expression.

F.3 Cutoff elasticities

Recall $\Delta \equiv 1 - d \log w_C / d\psi_v$. In the one-origin fixed-wage benchmark, (12) implies:

$$L(c_D - h e^{-\rho\psi_f/2})^{k+1} [c_D + (k+1)h e^{-\rho\psi_f/2}] e^{k\psi_v} = B.$$

Differentiating at $\psi_v = \psi_f = 0$ gives

$$\varepsilon_D^v = \frac{k(c_D - h)[c_D + (k+1)h]}{(k+2)c_D(c_D + kh)}, \quad \varepsilon_D^f = \frac{\rho(k+1)h^2}{2c_D(c_D + kh)}. \quad (32)$$

The limits are useful. If $h \rightarrow 0$, then $\varepsilon_D^v \rightarrow k/(k+2)$ and $\varepsilon_D^f \rightarrow 0$. Fixed-cost support has no fixed-cost channel when there is no fixed cost.

In the multi-origin case, cutoff responses solve a linear system. Define

$$\Pi_{on} \equiv L_n \tau_{on}^{-k} (c_{Dn} - h_{on})^{k+1} [c_{Dn} + (k+1)h_{on}], \quad \omega_{on} \equiv \frac{\Pi_{on}}{\sum_m \Pi_{om}}.$$

Let

$$K_{on} \equiv \frac{(c_{Dn} + kh_{on})c_{Dn}}{(c_{Dn} - h_{on})[c_{Dn} + (k+1)h_{on}]}, \quad J_{Cn} \equiv \frac{h_{Cn}^2}{(c_{Dn} - h_{Cn})[c_{Dn} + (k+1)h_{Cn}]}.$$

For the square case with the same number of origins and destinations, stack rows by origin and columns by destination and define $M_{on} \equiv \omega_{on} K_{on}$. The cutoff responses $x_n^v = d \log c_{Dn} / d\psi_v$ and $x_n^f = d \log c_{Dn} / d\psi_f$ solve

$$\mathbf{M}x^v = \begin{pmatrix} -k\Delta/(k+2) \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{M}x^f = \begin{pmatrix} -\frac{\rho(k+1)}{2} \sum_m \omega_{Cm} J_{Cm} \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

If $\mathbf{M}$ is nonsingular, Cramer's rule gives

$$\varepsilon_{Dn}^v = \frac{k}{k+2} \frac{\mathcal{C}_{C,n}(\mathbf{M})}{\det \mathbf{M}}, \quad \varepsilon_{Dn}^f = \frac{\rho(k+1)}{2} \sum_m \omega_{Cm} J_{Cm} \frac{\mathcal{C}_{C,n}(\mathbf{M})}{\det \mathbf{M}}, \quad (33)$$

where $\mathcal{C}_{C,n}(\mathbf{M})$ is the (C, n) cofactor. This expression motivates using China's pre-policy market share as a reduced-form exposure proxy for destinations where cutoff compression from Chinese policy is larger. The empirical design does not estimate the cofactor.

F.4 Chinese comparative statics

For prices and markups, differentiate (5). Using (14), $d \log c_{in} / d\psi_v = -\Delta$, and $d \log c_{in} / d\psi_f = 0$ yields (15)–(18).

For revenue, differentiate $r_{in} = L_n(c_{Dn}^2 - c_{in}^2)/(4\gamma)$:

$$\frac{d \log r_{in}}{d\psi_v} = \frac{2\Delta(c_{in}^2 - \varepsilon_{Dn}^v c_{Dn}^2)}{c_{Dn}^2 - c_{in}^2}, \quad \frac{d \log r_{in}}{d\psi_f} = -\frac{2\varepsilon_{Dn}^f c_{Dn}^2}{c_{Dn}^2 - c_{in}^2}.$$

The variable-cost component increases incumbent revenue iff $c_{in}/c_{Dn} > \sqrt{\varepsilon_{Dn}^v}$. The fixed-cost component lowers incumbent revenue in markets with $\varepsilon_{Dn}^f > 0$ because cutoff compression reduces residual demand.

For destination scope,

$$\frac{d \log(c_{in}/\tilde{c}_{Dn}^C)}{d\psi_v} = -\Delta + \frac{c_{Dn}\varepsilon_{Dn}^v \Delta}{\tilde{c}_{Dn}^C}, \quad \frac{d\tilde{c}_{Dn}^C}{d\psi_f} = \frac{\rho h_{Cn}}{2} - c_{Dn}\varepsilon_{Dn}^f.$$

The first expression is negative when $\varepsilon_{Dn}^v < \tilde{c}_{Dn}^C/c_{Dn}$. The second is positive when the direct reduction in the participation cost dominates the induced cutoff compression.

F.5 Aggregate revenue

Total Chinese revenue in destination n , holding the entrant mass fixed, is

$$R_{Cn} = \frac{M_{E,C} L_n (c_{Dn} - h_{Cn})^k N_{Cn}}{4\gamma(k+2)c_{\max,Cn}^k}, \quad N_{Cn} \equiv 2c_{Dn}^2 + 2kc_{Dn}h_{Cn} - kh_{Cn}^2.$$

Define

$$A_{Cn} \equiv \frac{c_{Dn} [2c_{Dn}^2 + 2(k-1)c_{Dn}h_{Cn} - kh_{Cn}^2]}{(c_{Dn} - h_{Cn})N_{Cn}}.$$

Then

$$\frac{d \log R_{Cn}}{d\psi_v} = \Delta [k - (k+2)A_{Cn}\varepsilon_{Dn}^v],$$

and

$$\frac{d \log R_{Cn}}{d\psi_f} = \frac{\rho k(k+2)h_{Cn}^2(2c_{Dn} - h_{Cn})}{2(c_{Dn} - h_{Cn})N_{Cn}} - (k+2)A_{Cn}\varepsilon_{Dn}^f.$$

When $h_{Cn} = 0$, $A_{Cn} = 1$ and the variable-cost expression reduces to $\Delta[k - (k+2)\varepsilon_{Dn}^v]$. Aggregating over destinations gives the reversal condition $\sum_n \omega_{Cn}^R \varepsilon_{Dn}^v < k/(k+2)$ in the no-fixed-cost benchmark.

F.6 French revenue and employment

French delivered cost c_{iF} is unaffected directly by Chinese policy. Differentiating $r_{iF,n} = L_n(c_{Dn}^2 - c_{iF}^2)/(4\gamma)$ gives

$$\frac{d \log r_{iF,n}}{d\psi_v} = -\nu_{iF,n} \varepsilon_{Dn}^v \Delta, \quad \frac{d \log r_{iF,n}}{d\psi_f} = -\nu_{iF,n} \varepsilon_{Dn}^f,$$

where $\nu_{iF,n} = 2c_{Dn}^2/(c_{Dn}^2 - c_{iF}^2) > 1$. Employment is

$$\ell_{iF,n} = \frac{L_n c_{iF} (c_{Dn} - c_{iF})}{2\gamma w_F},$$

so employment falls in markets where Chinese policy lowers c_{Dn} .

F.7 Input-output propagation

The input-output extension is an accounting layer on top of the sector-level comparative statics. The own-sector variable-cost component changes effective costs in sector s by $-\Delta_s \psi_{v,s}$. If sector s uses inputs from sector $\hat{s}$ with cost share $\alpha_{s\hat{s}}$, a policy-induced price change in sector $\hat{s}$ changes sector- s input costs by

$$\alpha_{s\hat{s}} \left(\pi_{\hat{s}}^v \psi_{v,\hat{s}} + \pi_{\hat{s}}^f \psi_{f,\hat{s}} \right),$$

where $\pi_{\hat{s}}^v$ and $\pi_{\hat{s}}^f$ are average pass-through coefficients in the Chinese market. Policy in sector $\hat{s}$ can also raise demand for inputs supplied by sector s . We summarize this demand-pressure

channel by

$$\lambda_s \alpha_{\hat{s}s} (\psi_{v,\hat{s}} + \eta \psi_{f,\hat{s}}),$$

where λ_s maps demand expansion into input-cost pressure and $\eta \in [0, 1]$ allows fixed-cost support to generate weaker demand effects than variable-cost support. Summing the own-sector, downstream, and upstream terms gives (26).

G Additional results on Chinese firms

This section reports Chinese-firm outcomes that support the empirical interpretation in Section 6 but are not the primary model-implied margins. The event-study specification is the same staggered provincial policy-onset design used in Sections 4 and 6. The figure documents domestic adjustment; the main text uses export sales, destination scope, quality-adjusted prices, and markups as the model tests.

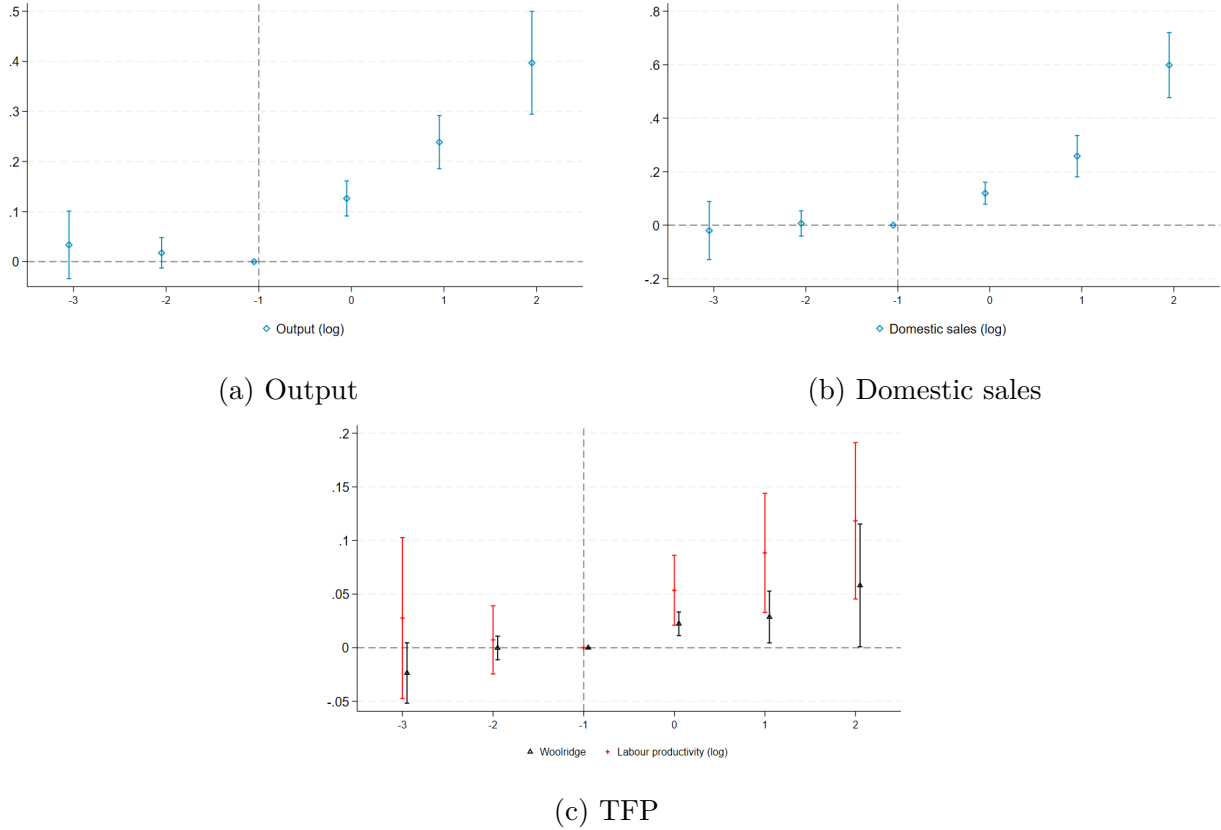


Figure 12: Dynamic effects of provincial industrial policy on firm performance

Notes: Fully interacted cohort-by-event-time DiD estimator following Sun and Abraham (2021). Never-treated sector-province cells form the comparison group. Standard errors clustered at the sector-province level.

Product scope is a diagnostic for the export expansion in Table 2. The model's market-access margin is destination participation, not the number of HS products shipped by a firm. Estimating the same PPML specification with product count as the outcome shows whether the export-value response reflects broader product scope. The estimates are small and im-

precise, so the main export response is best read as higher sales and broader destination reach within existing product lines.

Table 15: Industrial policy and export product scope

	# products	
	(1)	(2)
Panel A. IP score	−0.095 (0.063)	−0.066 (0.053)
Panel B. Funding score	−0.596 (0.426)	−0.530 (0.377)
Panel C. Promotion score	−0.105 (0.071)	−0.072 (0.060)
Observations	1,160,871	1,160,871
Firm FE	Yes	Yes
Year FE	Yes	No
Province × year FE	No	Yes

Notes: PPML regressions. Each panel reports a separate regression. Standard errors are clustered at the sector-province level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The full Broad Economic Category split extends Table 3. The main text focuses on intermediate and final goods because those groups map most directly to input costs and consumer-market competition. Primary and capital goods are included here to show the full pattern of heterogeneity. The negative estimates for capital and final goods and the positive estimates for primary goods indicate that price responses differ across product classes; this heterogeneity should not be interpreted as a separate prediction of the benchmark model.

Table 16: Industrial policy and quality-adjusted export prices: Broad Economic Categories

	Log quality-adjusted unit value				
	All goods (1)	Primary goods (2)	Intermediate goods (3)	Capital goods (4)	Final goods (5)
Panel A. IP score	−0.044 (0.031)	0.150*** (0.056)	−0.043 (0.058)	−0.384*** (0.097)	−0.267*** (0.062)
Panel B. Funding score	−0.632** (0.290)	1.708*** (0.586)	−0.633 (0.511)	−2.246*** (0.848)	−1.583*** (0.520)
Panel C. Promotion score	−0.042 (0.033)	0.160*** (0.061)	−0.039 (0.064)	−0.426*** (0.106)	−0.300*** (0.070)
Observations	862,330	85,904	673,237	429,558	532,455
Firm FE	Yes	Yes	Yes	Yes	Yes
Province × year FE	Yes	Yes	Yes	Yes	Yes

Notes: OLS regressions. Each panel reports a separate regression. Quality adjustment follows [Khandelwal et al. \(2013\)](#). Standard errors are clustered at the sector-province level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

G.1 Input-output price propagation

Proposition 5 shows that input-output linkages have an ambiguous sign. Policy in an upstream sector can lower downstream input costs, while expansion in a downstream sector can raise demand for upstream inputs. The empirical object here is the quality-adjusted export price of Chinese firms, regressed on IO-weighted policy exposure in linked sectors. The estimate is therefore a diagnostic for indirect price propagation inside China, not an input to the French displacement equation. Table 17 reports the all-goods specification. The coefficient on total IO exposure is negative, which indicates that linked-sector policy exposure is associated with lower quality-adjusted prices on average.

Table 17: Input-output exposure and quality-adjusted export prices

	Log quality-adjusted unit value
	All goods (1)
Panel A. Total IO exposure	−0.049** (0.025)
Panel B. Funding IO exposure	−0.706*** (0.259)
Panel C. Promotion IO exposure	−0.048* (0.026)
Observations	861,998
Firm FE	Yes
Province × year FE	Yes

Notes: OLS regressions. The outcome is the log quality-adjusted export unit value. Input-output exposure weights policy in linked sectors by input-output shares. Quality adjustment follows [Khandelwal et al. \(2013\)](#). Standard errors are clustered at the sector-province level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

H Markup estimation details

Markup formula. Following [De Loecker and Warzynski \(2012\)](#), a cost-minimizing firm’s markup can be recovered from the output elasticity of a flexible input divided by that input’s revenue share. Using materials as the flexible input,

$$\mu_{it} = \theta_{it}^M \frac{P_{it}Q_{it}}{P_{it}^M M_{it}}, \quad (34)$$

where μ_{it} is the markup, θ_{it}^M is the output elasticity of materials, $P_{it}Q_{it}$ is revenue, and $P_{it}^M M_{it}$ is materials expenditure.

Data construction. The NBS database reports gross output, employment, net fixed assets, total wages, cost of main business, current-year depreciation, and input VAT. Intermediate input expenditures are not directly reported for the 2011–2014 surveys. The baseline construction is $M_{it} = \text{cost of main business}_{it} - W_{it} - \text{depreciation}_{it}$, which removes labor

and capital costs from the aggregate cost measure. All nominal variables are deflated using 2-digit industry-level output and input price indices from the NBS.

Production function. We estimate a Cobb-Douglas gross-output production function separately for each 2-digit CIC industry:

$$y_{it} = \beta_l l_{it} + \beta_k k_{it} + \beta_m m_{it} + \zeta_{it} + \epsilon_{it}, \quad (35)$$

where lowercase letters denote logs and ζ_{it} is unobserved productivity. Under Cobb-Douglas, β_m is the industry-specific output elasticity of materials. We estimate (35) using the control-function approach of [Akerberg et al. \(2015\)](#): materials demand proxies for unobserved productivity, capital and labor are predetermined, and materials are flexible. We use Cobb-Douglas rather than translog because the 2011–2014 panel leaves few effective lags for estimating interaction terms.

Given $\hat{\beta}_m$, the firm-level markup is:

$$\hat{\mu}_{it} = \hat{\beta}_m \frac{P_{it} Q_{it}}{P_{it}^M M_{it}}. \quad (36)$$

Markups are winsorized at the 1st and 99th percentiles within industry.

Additional markup regressions. The first diagnostic estimates within-firm markup correlations using the same policy index and fixed-effect structure as the main Chinese-firm regressions. The second allows the policy-markup relationship to vary with the firm’s export share. These exercises check whether the event-study null in Section 6.3 hides systematic markup differences by exposure or export orientation. They are not used as the preferred causal evidence.

Table 18: Industrial policy and firm-level markups: cross-sectional correlations

	Log markup					
	All firms		Domestic-oriented		Exporters	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. IP score ($\times 100$)	0.029*** (0.009)	0.029*** (0.009)	0.041*** (0.010)	0.040*** (0.009)	0.019 (0.013)	0.021 (0.013)
Panel B. Funding score ($\times 100$)	0.233*** (0.084)	0.260*** (0.088)	0.304*** (0.072)	0.333*** (0.076)	0.173 (0.115)	0.205* (0.111)
Panel C. Promotion score ($\times 100$)	0.029*** (0.010)	0.029*** (0.009)	0.041*** (0.012)	0.040*** (0.010)	0.019 (0.013)	0.020 (0.014)
Observations	697,623	697,623	307,530	307,530	390,093	390,093
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes	No
Province $\times$ year FE	No	Yes	No	Yes	No	Yes

Notes: OLS regressions. Markups are constructed using (36). Domestic-oriented firms have zero customs export value. Exporters have positive customs export value. Standard errors are clustered at the sector-province level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 19: Industrial policy and firm-level markups: export-share interactions

	Log markup		
	IP score (1)	Funding score (2)	Promotion score (3)
Policy score ($\times 100$)	0.038*** (0.011)	0.340*** (0.110)	0.037*** (0.012)
Export share	-0.026*** (0.006)	-0.027*** (0.006)	-0.026*** (0.006)
Policy score $\times$ export share	-0.054* (0.030)	-0.462** (0.225)	-0.054 (0.033)
Observations	436,229	436,229	436,229
Firm FE	Yes	Yes	Yes
Province $\times$ year FE	Yes	Yes	Yes

Notes: OLS regressions. Export share is customs export value divided by total sales. Standard errors are clustered at the sector-province level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

I Additional results on French firms

This section contains secondary French outcomes. They are not the primary margins in Propositions 3 and 4, but they discipline the interpretation of the firm-level displacement results.

The wage-bill regression uses the same firm-level exposure measure and fixed effects as Table 5. The outcome is the log wage bill. This margin helps separate the employment result from broader changes in labor payments and firm scale. The total-index coefficient is close to zero, while funding exposure is negative, mirroring the stronger production and employment responses to fiscal-support language in the main table.

The import regression examines whether exposed French firms offset export losses by sourcing more from China in the same products. This is a sourcing diagnostic rather than a displacement test. If cheaper Chinese products became inputs for French firms, imports from China could rise even as French exports fall. The total-index estimates are close to zero, so the main export displacement is not accompanied by a broad increase in French firm-product imports from China in this specification.

Table 20: Chinese industrial policy exposure and French firm wage bills

	Wage bill (1)
Export exposure: IP score	0.027 (0.064)
Export exposure: funding	−0.800** (0.374)
Export exposure: promotion	0.056 (0.077)
Observations	426,818
R^2	0.969
Firm FE	Yes
Year FE	Yes

Notes: OLS regression on French exporting firms. The outcome is the log wage bill. Exposure is defined in (30). Standard errors are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 21: Chinese industrial policy and French firm-product imports from China

	Import value		
	IP score (1)	Funding score (2)	Promotion score (3)
Panel A. National	−0.507* (0.302)	−1.349 (1.348)	−0.588* (0.345)
Panel B. Total	−0.061 (0.626)	0.042 (2.403)	−0.072 (0.700)
Observations	5,761,601	5,761,601	5,761,601
Firm $\times$ year FE	Yes	Yes	Yes
Product FE	Yes	Yes	Yes

Notes: PPML regressions. The outcome is French firm-product import value from China. Each column in each panel reports a separate regression. Standard errors are clustered at the Chinese industry level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.